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(54) **GLASS LAMINATE INCLUDING REFLECTIVE FILM**

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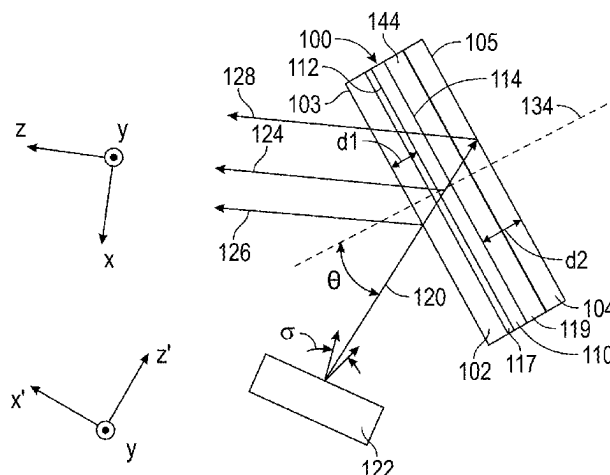
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(57) **ABSTRACT**

A glass laminate including first and second glass layers, a reflective film having opposed first and second major surfaces and disposed between the first and second glass layers with the first and second major surfaces facing the respective first and second glass layers, a first adhesive layer disposed between and bonding together the first glass layer and the

(Continued)



reflective film, and a second adhesive layer disposed between and bonding together the second glass layer and the reflective film is described. The second adhesive layer is thicker than the first adhesive layer such that the first major surface of the reflective film is separated from an outermost major surface of the first glass layer by a distance d1, the second major surface of the reflective film is separated from an outermost major surface of the second glass layer by a distance d2, and $0.05 \leq d1/d2 \leq 0.9$.

5 Claims, 16 Drawing Sheets

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G02B 27/00 (2006.01)

(52) U.S. Cl.

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(58) Field of Classification Search

USPC 359/485.01
See application file for complete search history.

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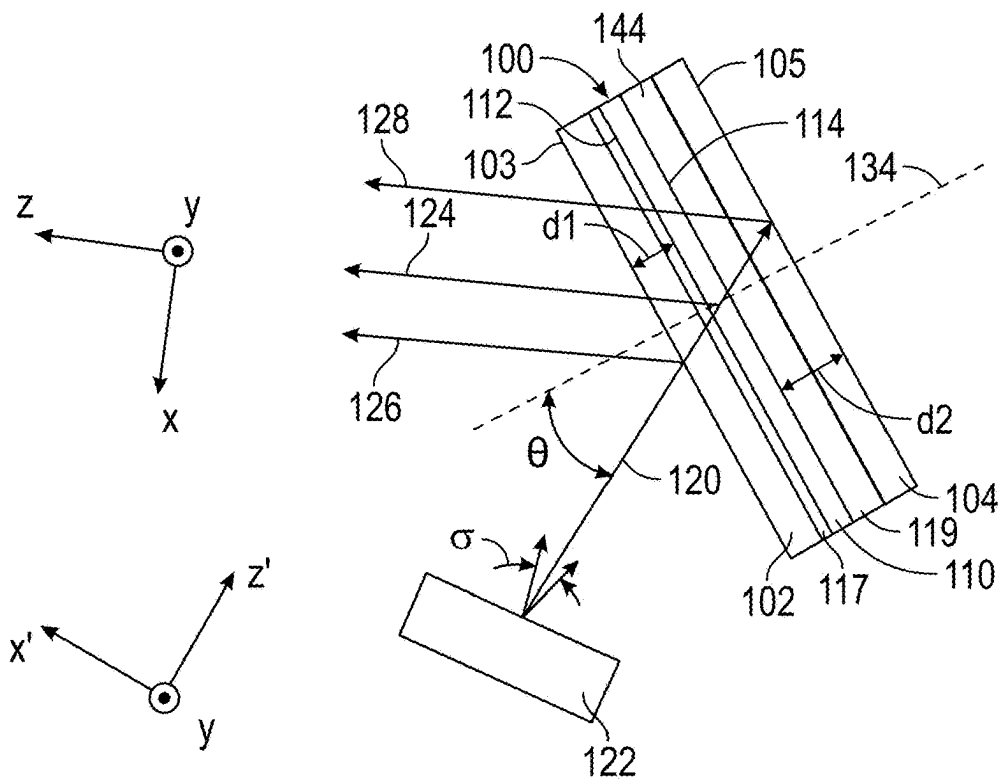


FIG. 1A

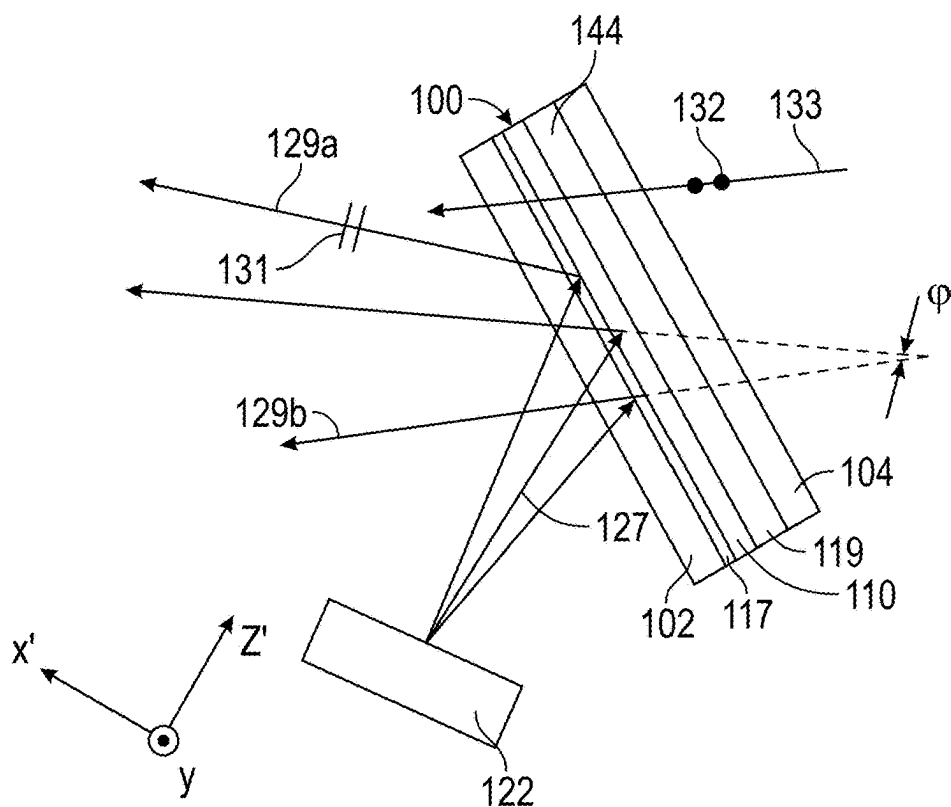


FIG. 1B

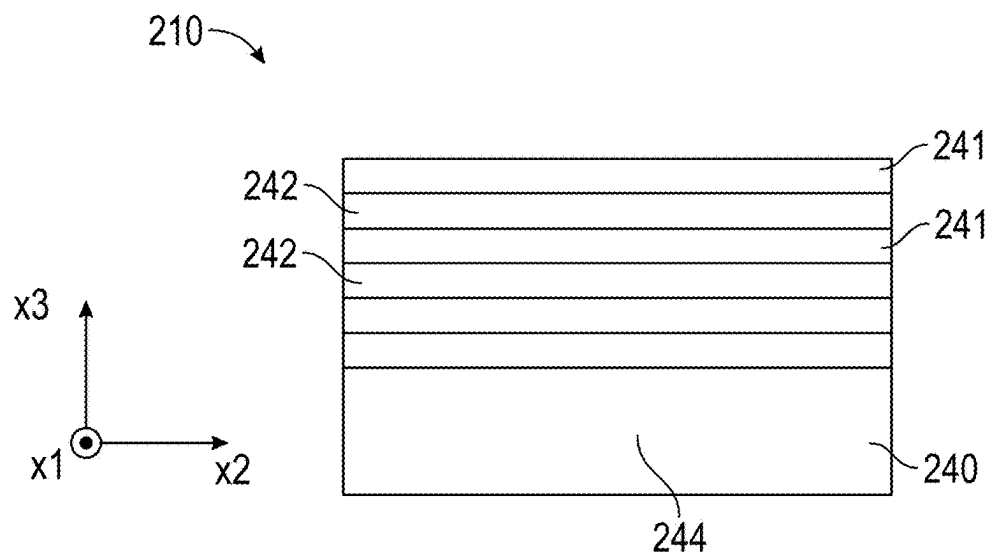


FIG. 2

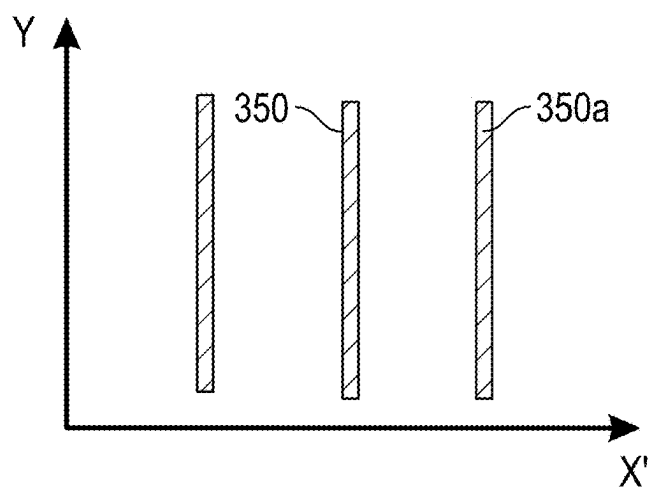


FIG. 3A

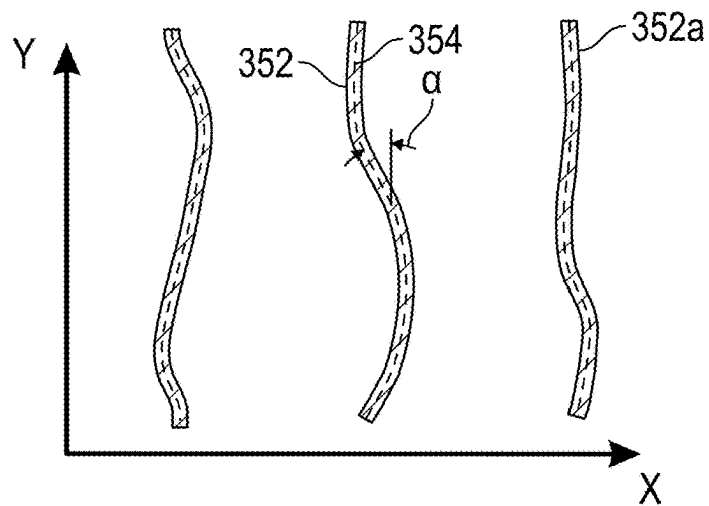


FIG. 3B

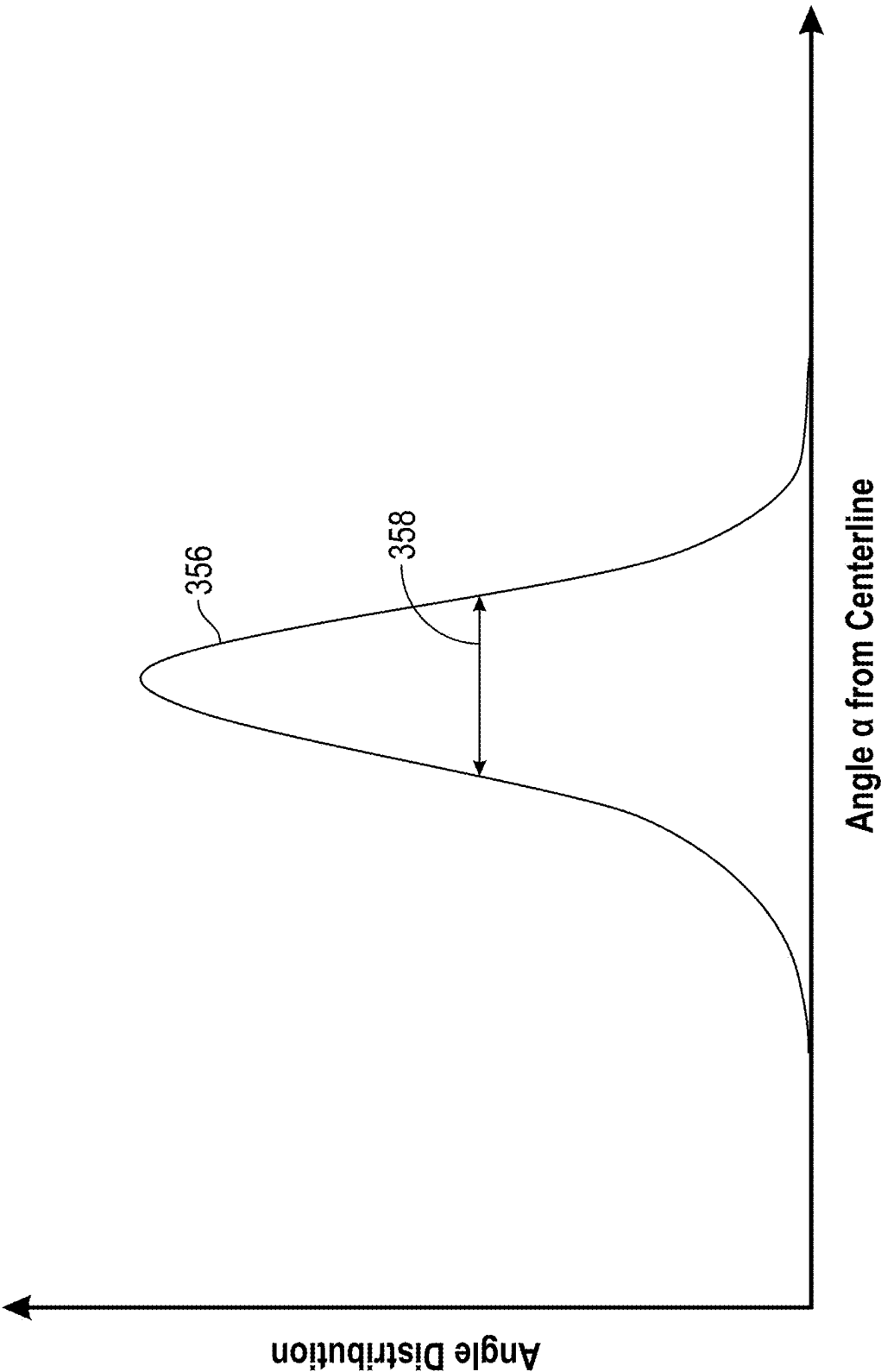


FIG. 3C

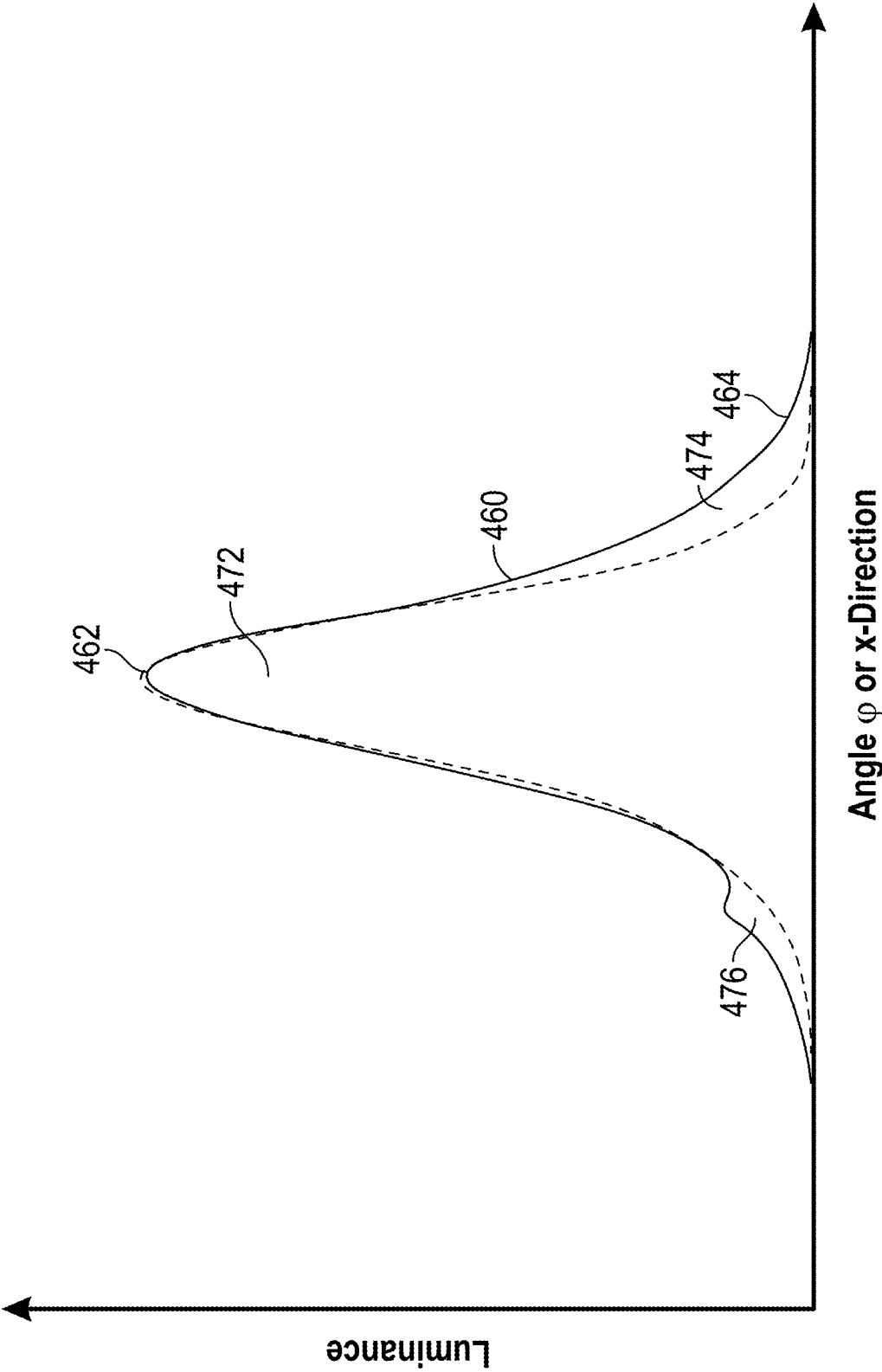


FIG. 4

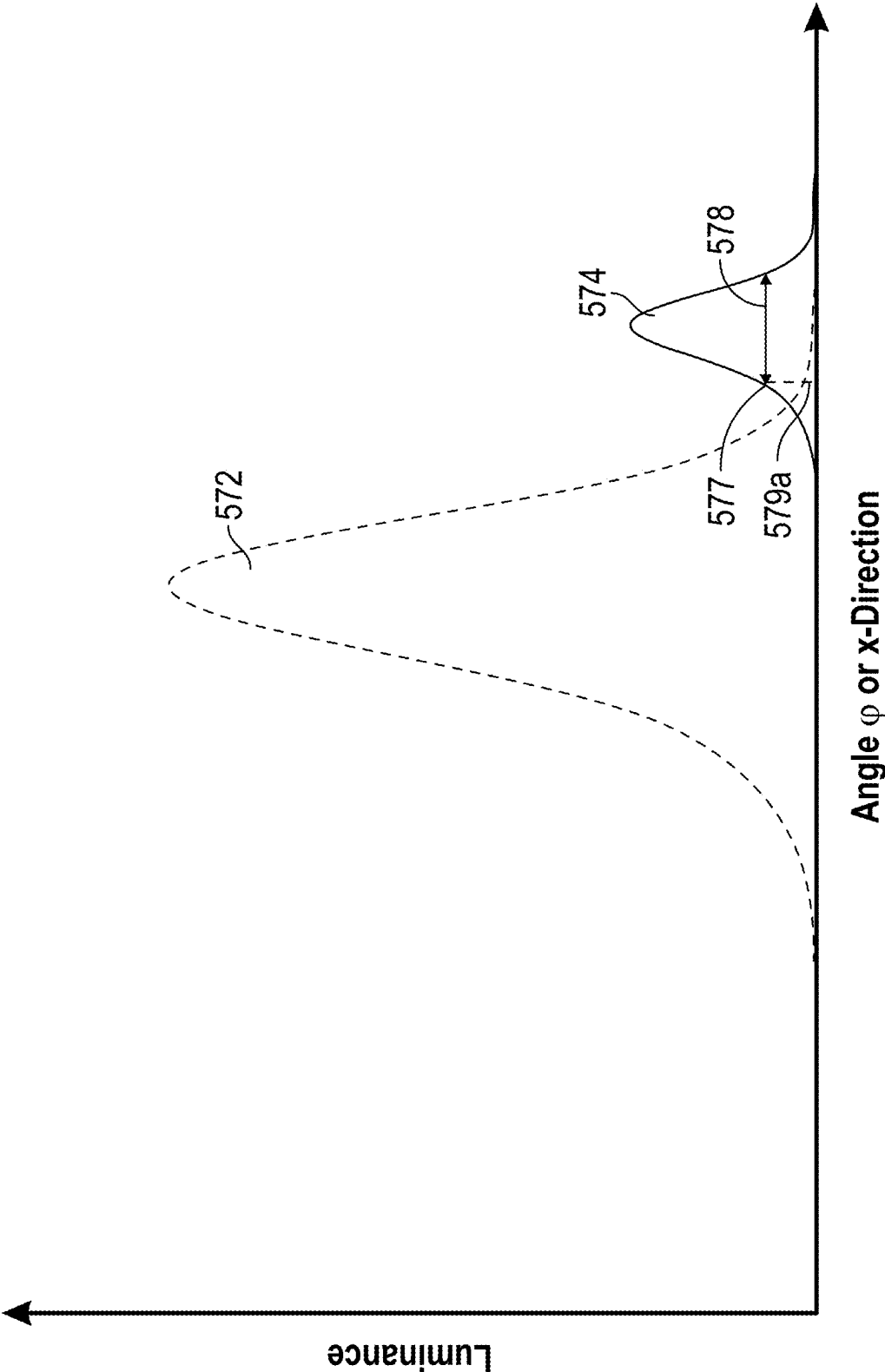
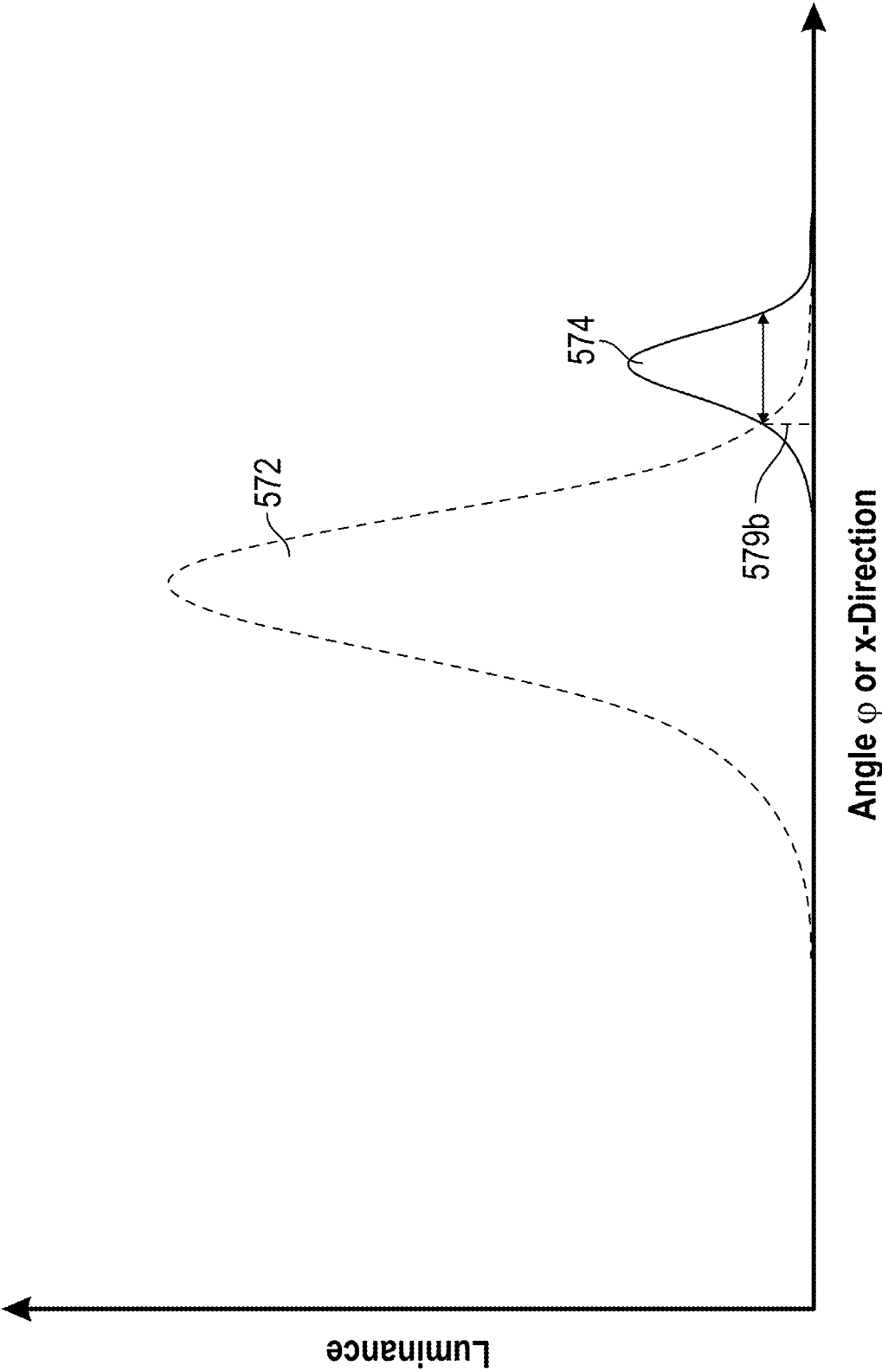
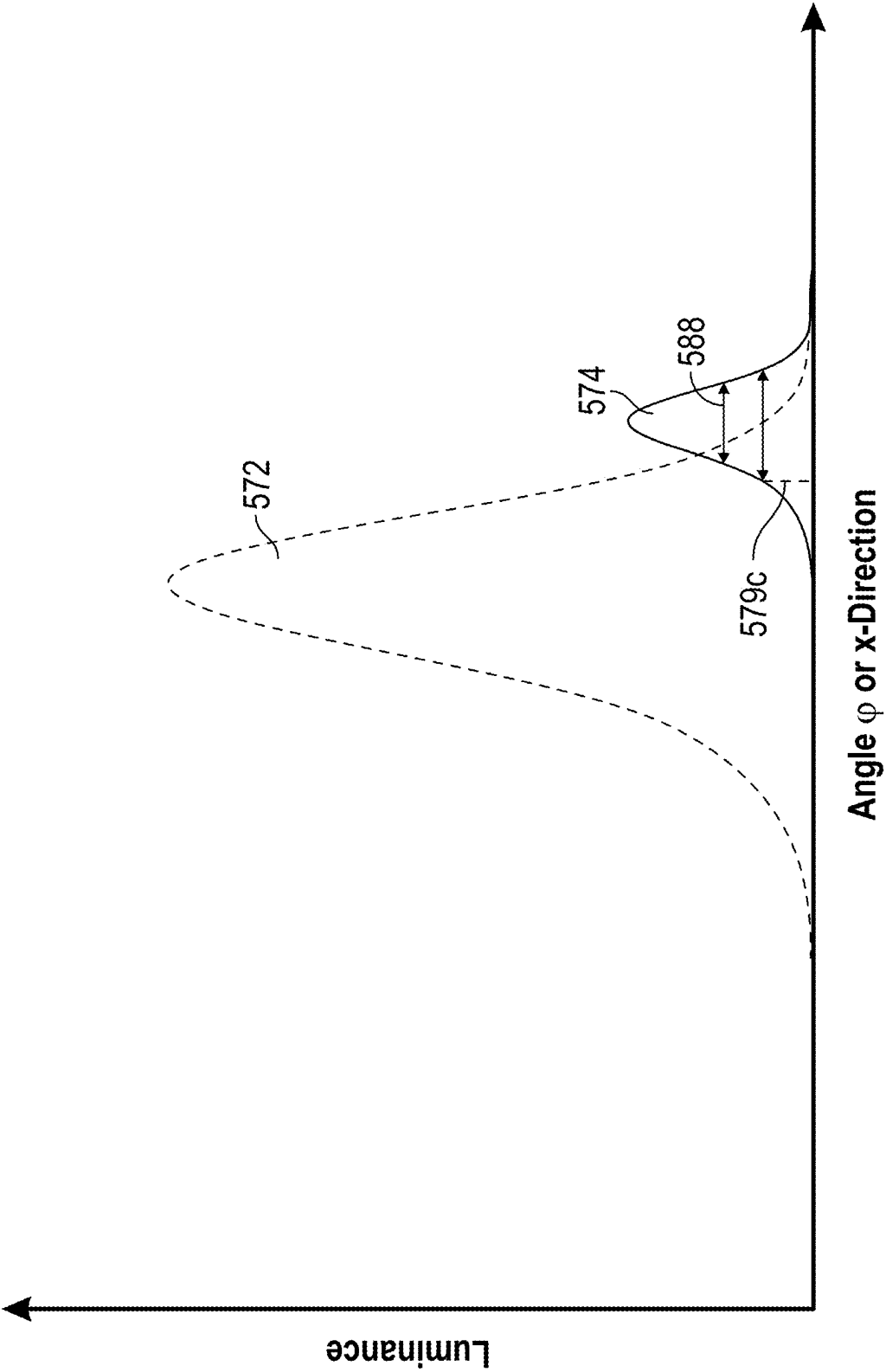


FIG. 5A



Angle φ or x-Direction

FIG. 5B



Angle ϕ or x-Direction

FIG. 5C

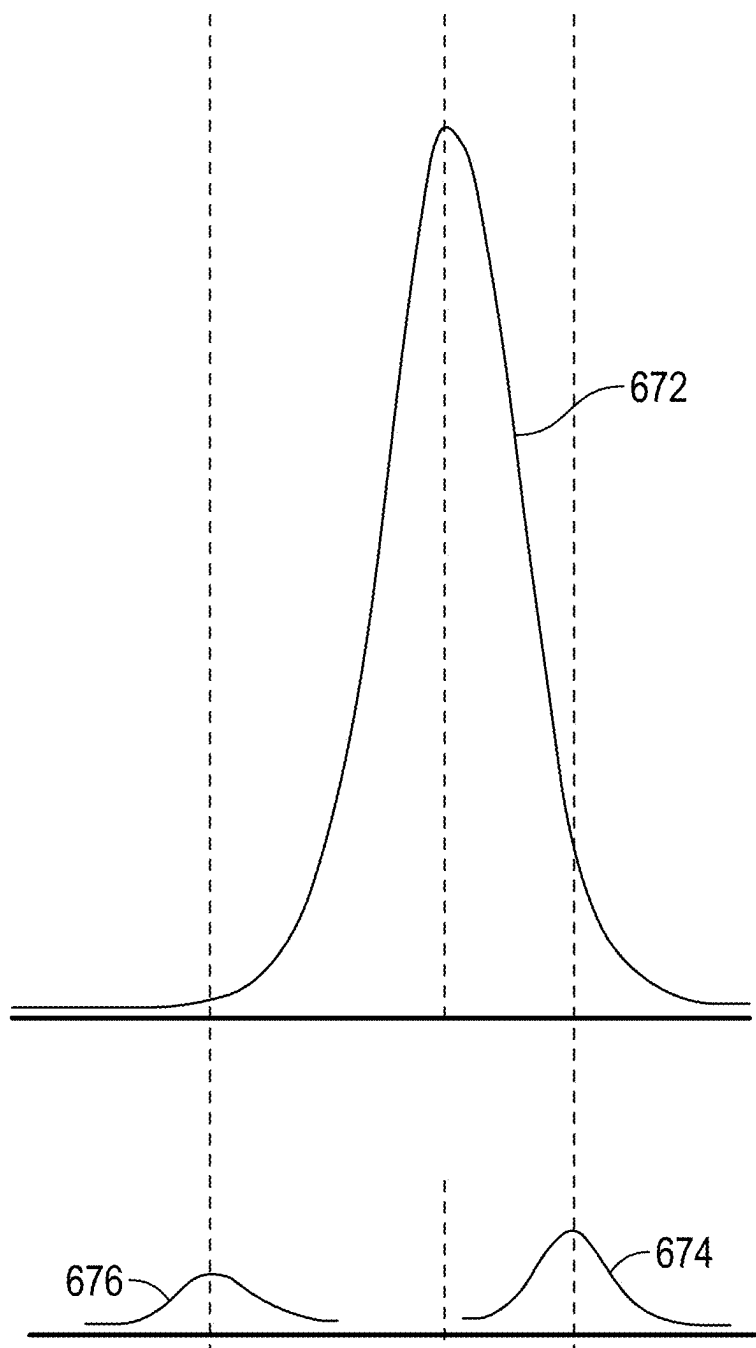


FIG. 6A

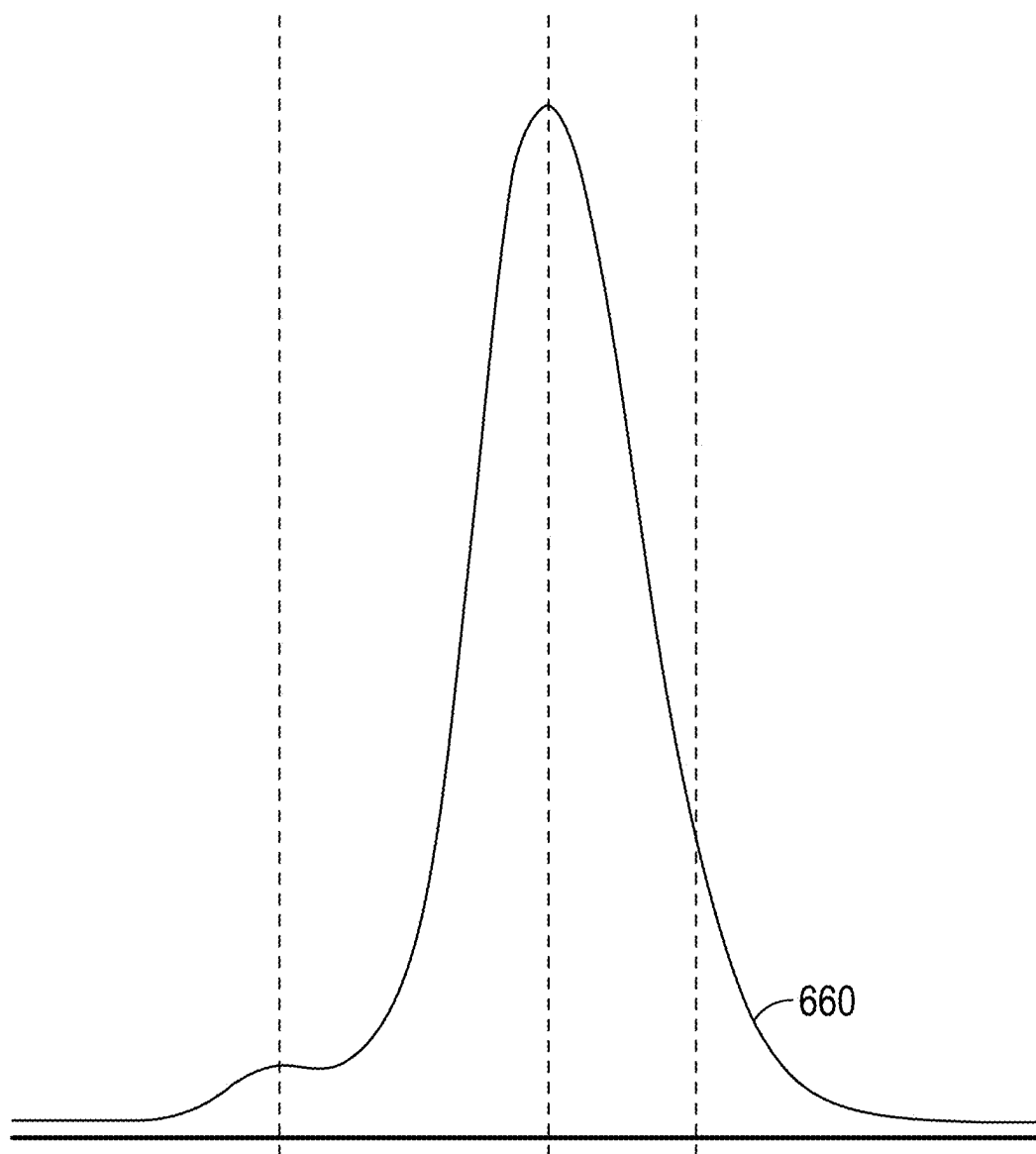


FIG. 6B

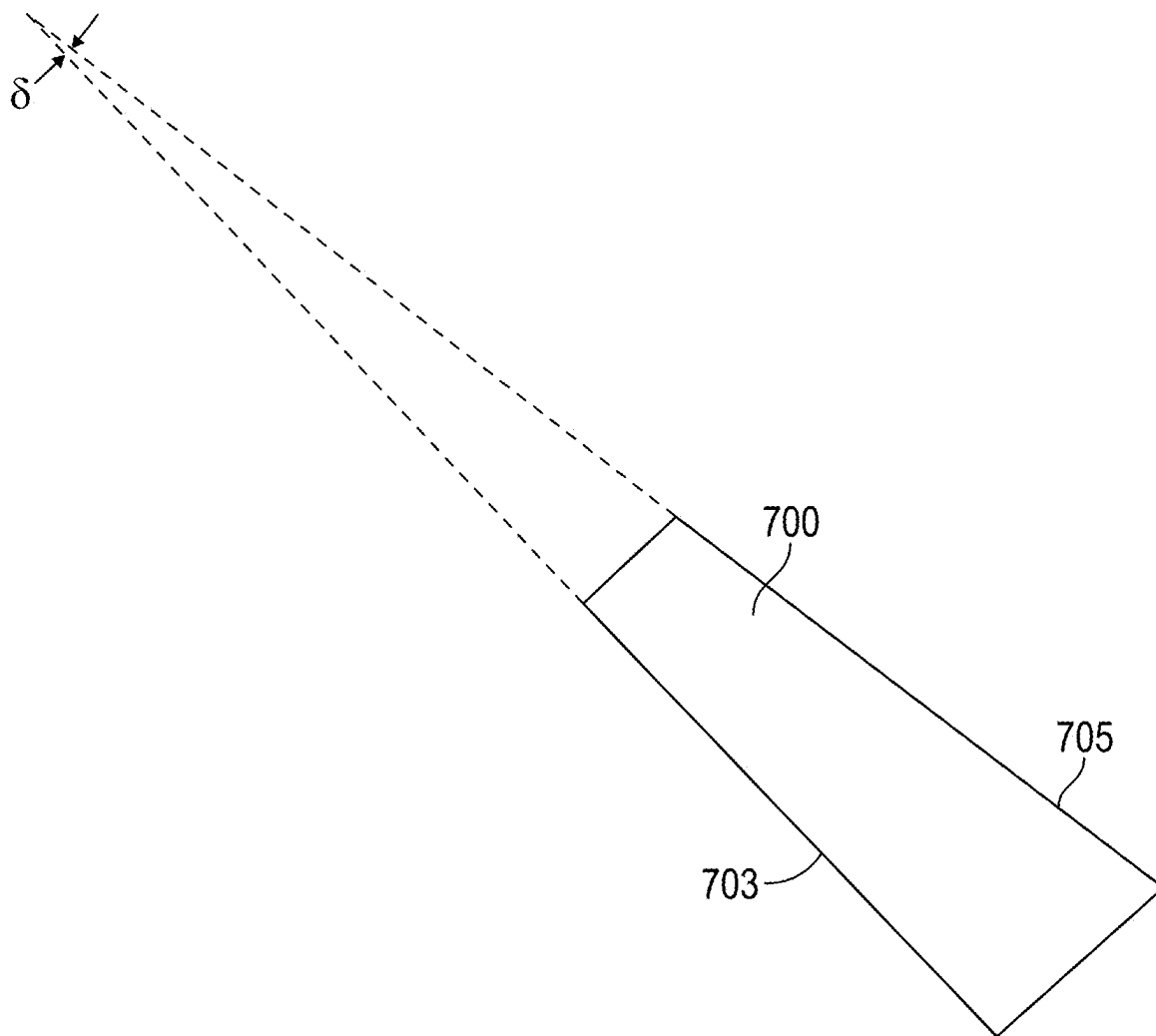


FIG. 7

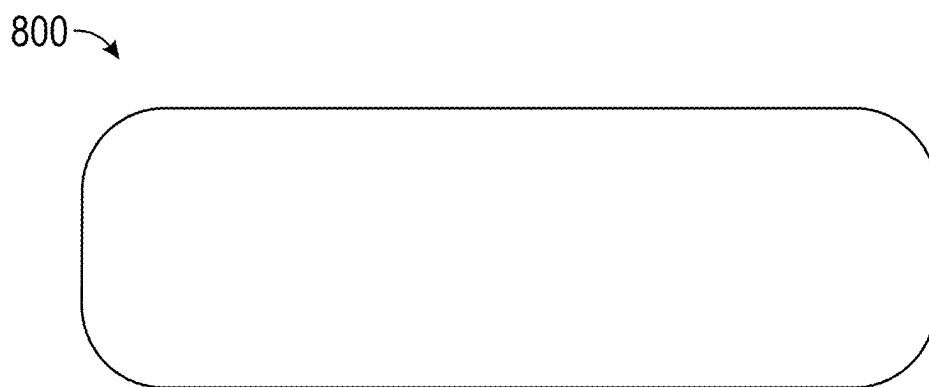


FIG. 8

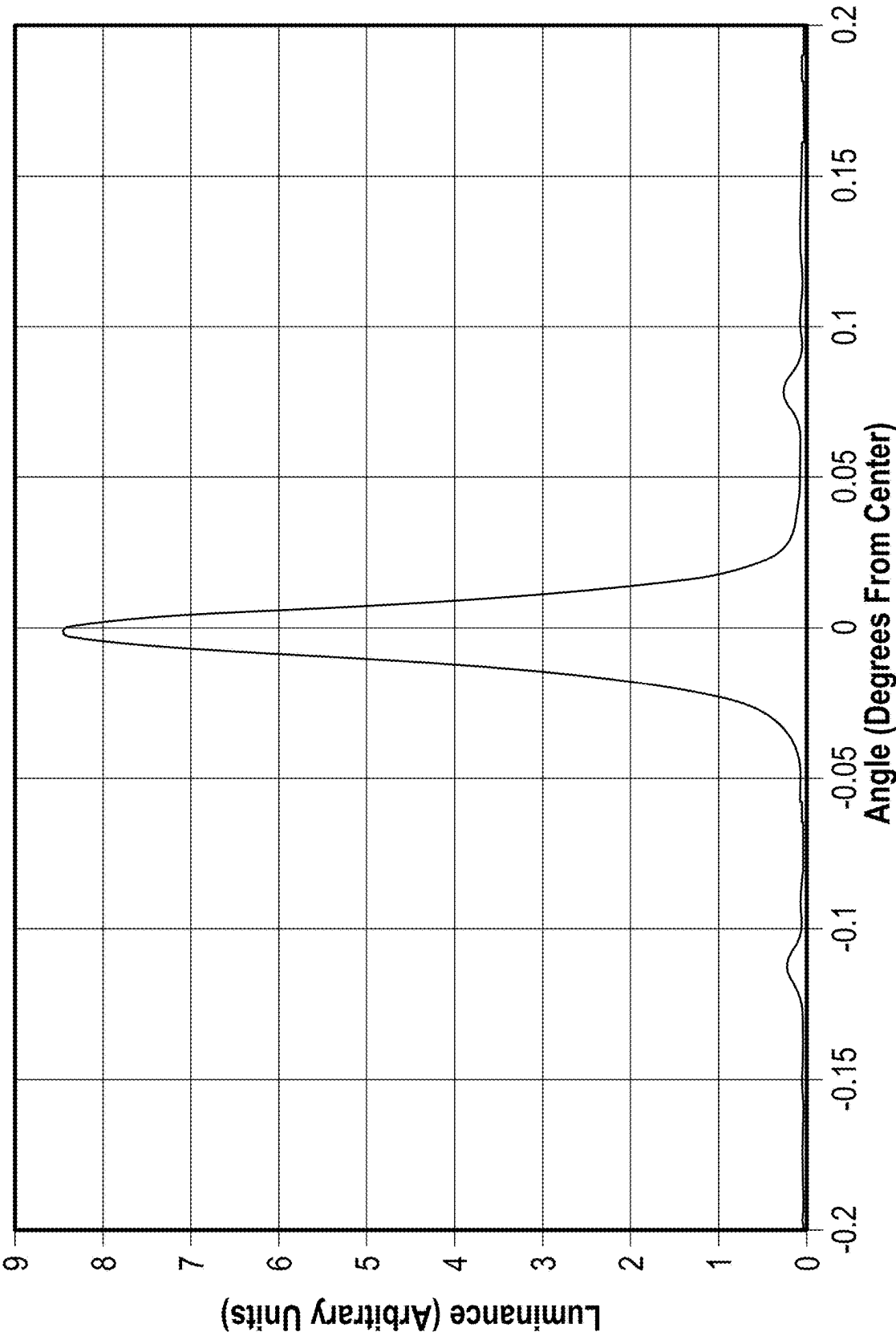
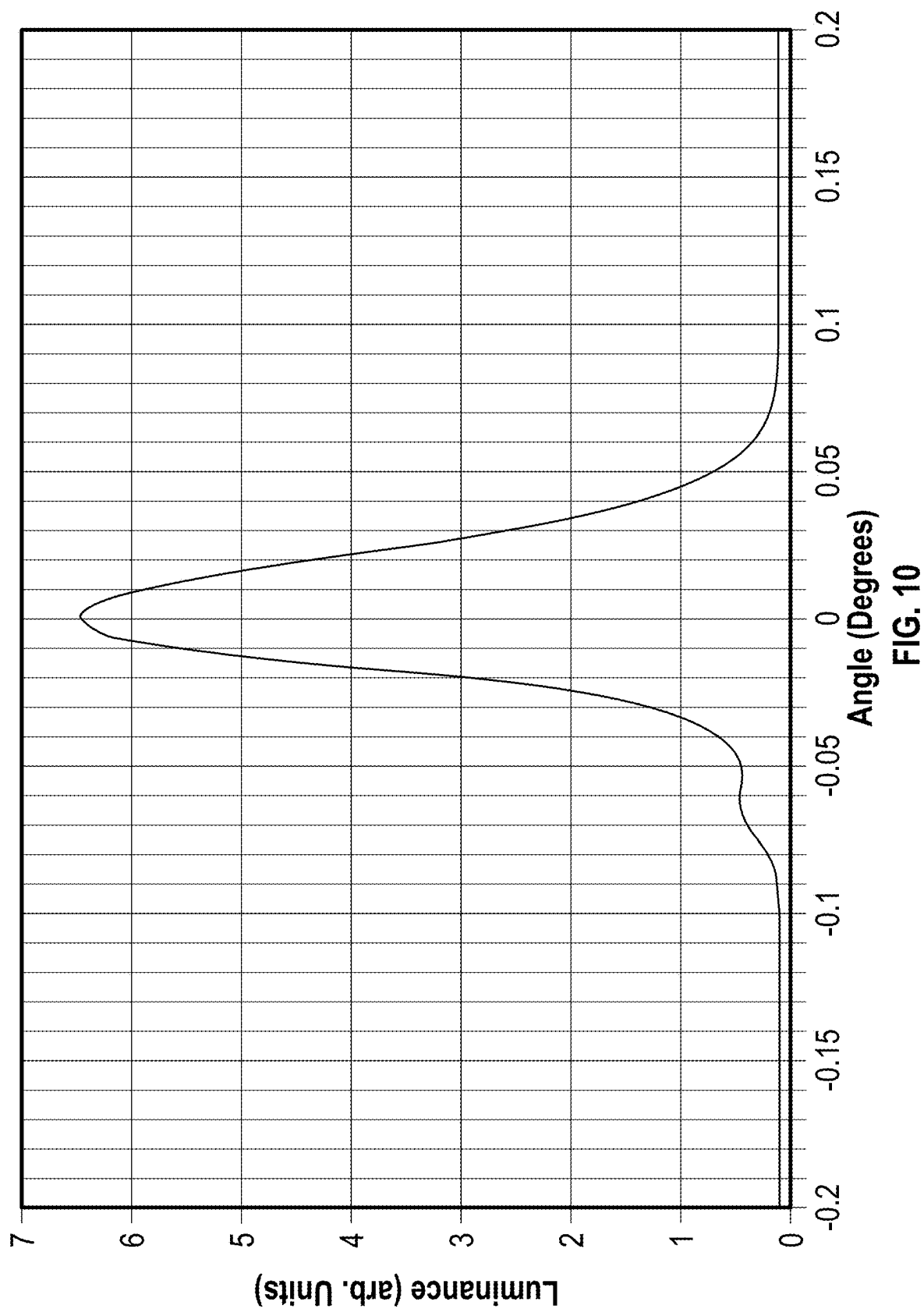


FIG. 9



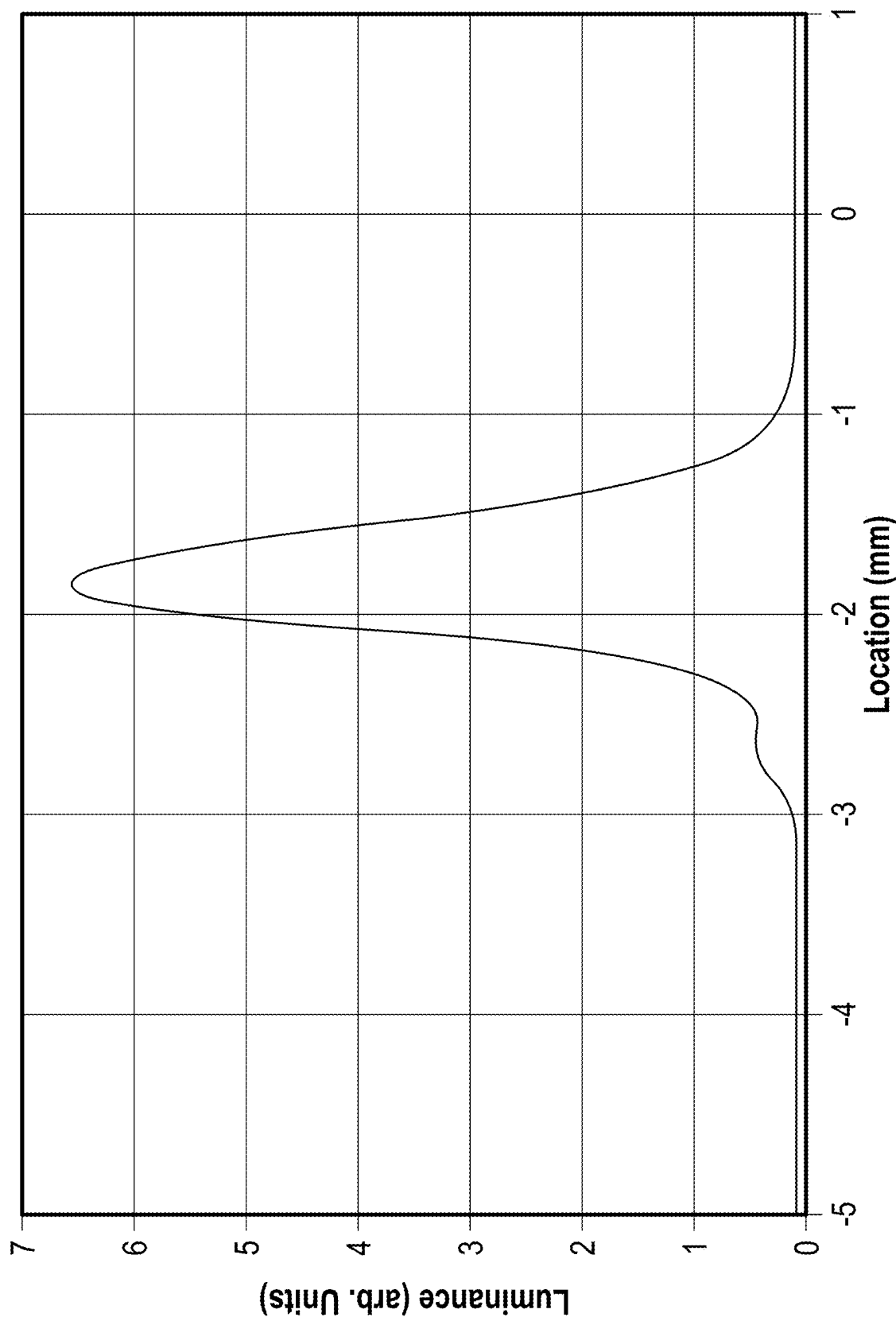


FIG. 11

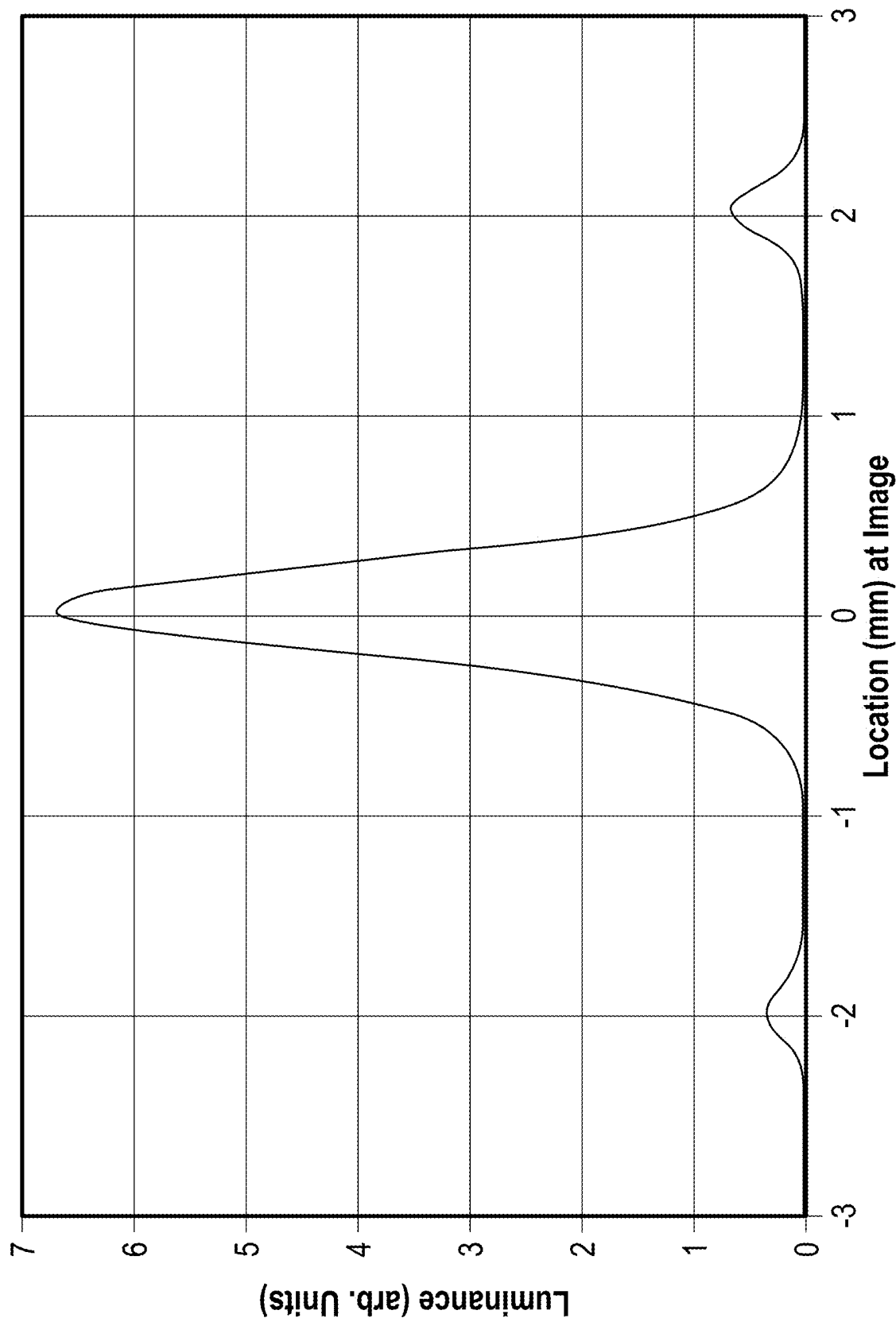


FIG. 12

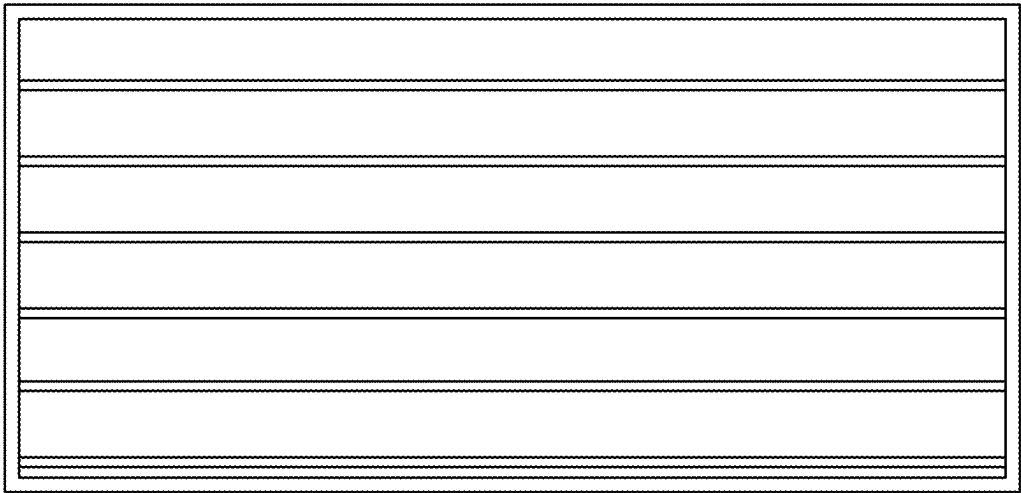


FIG. 13C

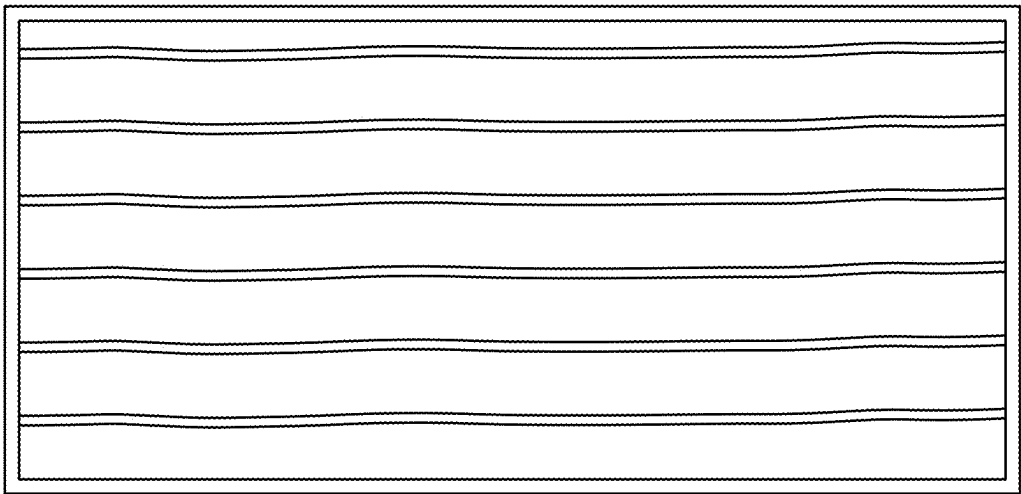


FIG. 13B

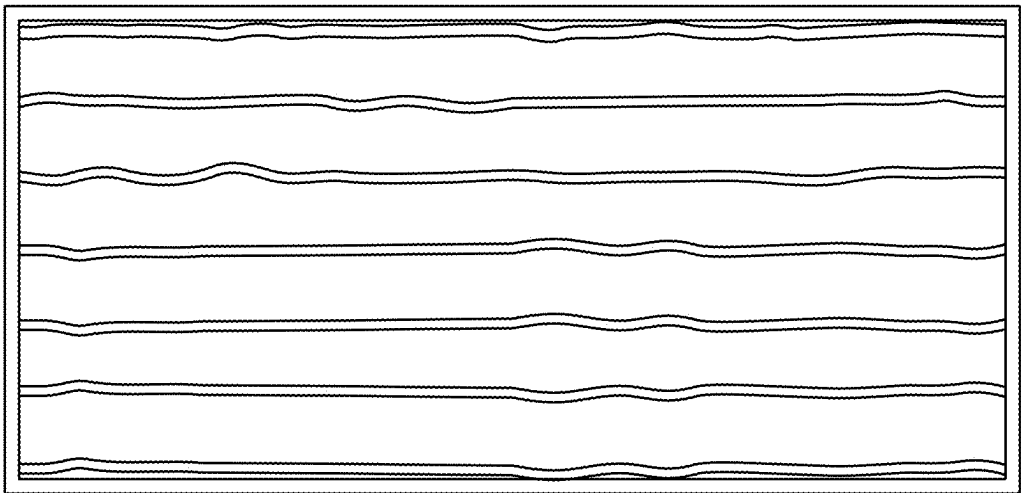


FIG. 13A

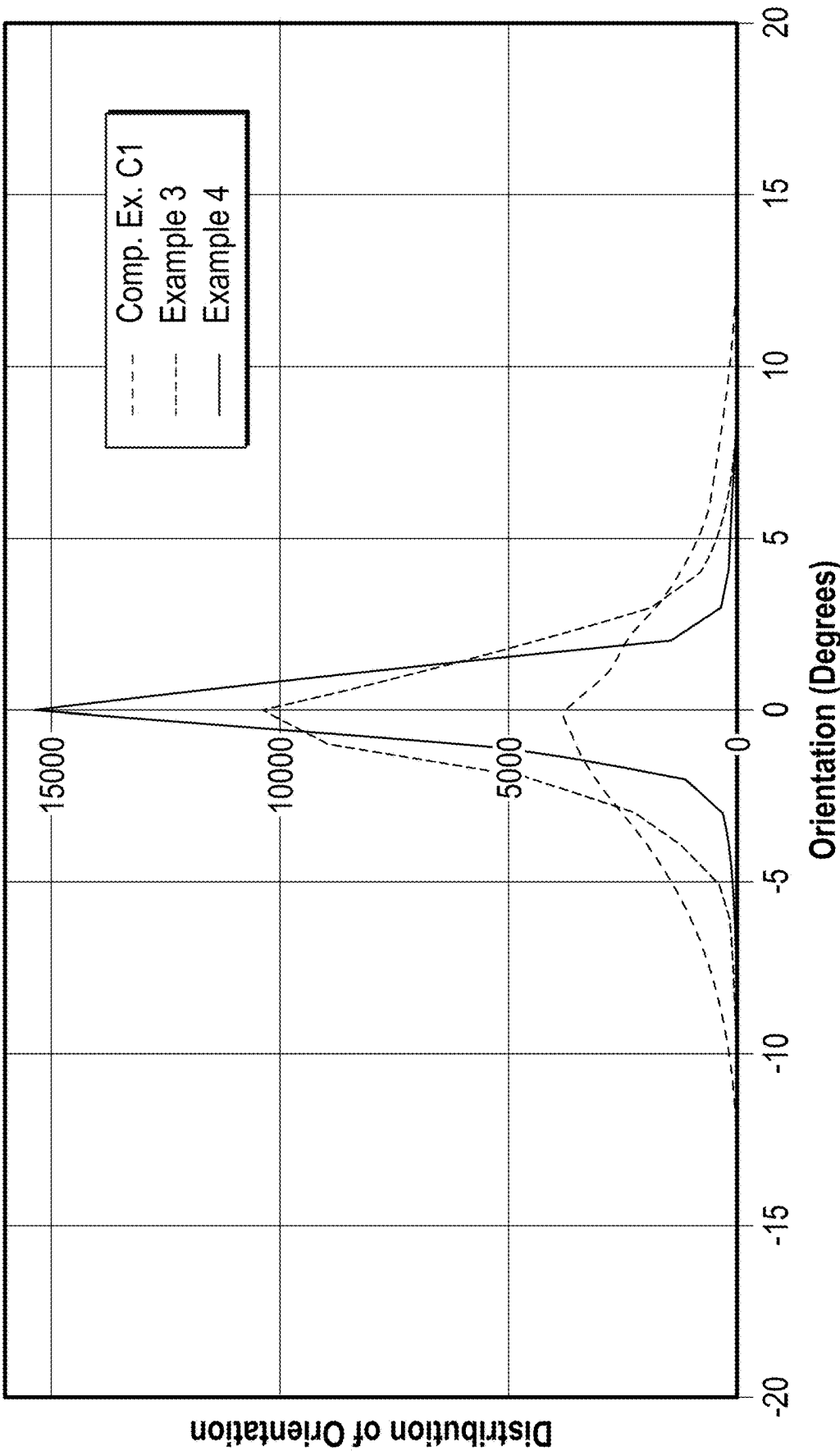


FIG. 14

GLASS LAMINATE INCLUDING REFLECTIVE FILM

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. application Ser. No. 17/274,594, filed Mar. 9, 2021, now allowed, which is a US 371 Application based on US2019/051733, filed on Sep. 18, 2019, which claims the benefit of U.S. Provisional Application No. 62/735,567, filed Sep. 24, 2018, the disclosures of which are incorporated by reference in their entireties herein.

BACKGROUND

Head-up displays may include a projector to project an image onto a windshield which reflects the projected image to a viewer. In some cases, the windshield is a glass laminate.

SUMMARY

In some aspects of the present description, a glass laminate including first and second glass layers having substantially parallel outermost major surfaces facing away from one another, a reflective film having opposed first and second major surfaces and disposed between the first and second glass layers with the first and second major surfaces facing the respective first and second glass layers, a first adhesive layer disposed between and bonding together the first glass layer and the reflective film, and a second adhesive layer disposed between and bonding together the second glass layer and the reflective film is provided. The reflective film has an average reflectance for a first polarization state in a predetermined visible wavelength range at a predetermined angle of incidence of at least 15% and an average transmittance for an orthogonal second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence of at least 30%. The second adhesive layer is thicker than the first adhesive layer such that the first major surface of the reflective film is separated from the outermost major surface of the first glass layer by a distance d_1 , the second major surface of the reflective film is separated from the outermost major surface of the second glass layer by a distance d_2 , and $0.05 \leq d_1/d_2 \leq 0.9$.

In some aspects of the present description, a glass laminate including first and second glass layers having substantially parallel outermost major surfaces, and a reflective film including a plurality of alternating polymeric interference layers and disposed asymmetrically between the outermost major surfaces is provided. When a light source positioned within 2 m of the glass laminate projects a line onto the outermost major surface of the first glass layer along a first direction making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal to the glass laminate so that the line extends along a second direction orthogonal to a first plane defined by the first direction and the normal and has a projected luminance distribution about a centerline of the projected line having a full width at half maximum of no more than 0.05 degrees, a first portion of the projected line reflects from the reflective film and a second portion of the projected line reflects from the outermost major surface of the first glass layer. A reflected image of the line includes a primary reflected image portion defined by the reflected first portion and a first ghost portion defined by the reflected

second portion. The first ghost portion substantially overlaps with the primary reflected image portion.

In some aspects of the present description, a glass laminate including first and second glass layers having substantially parallel outermost major surfaces, and a reflective film including a plurality of alternating polymeric interference layers and disposed between and adhered to the first and second glass layers through respective first and second adhesive layers is provided. The first adhesive layer has a thickness no more than 0.6 times a thickness of the second adhesive layer. When a light source projects a plurality of parallel lines onto the outermost major surface of the first glass layer along a first direction making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal to the glass laminate so that the plurality of parallel lines extend along a second direction orthogonal to a first plane defined by the first direction and the normal and are spaced apart along a third direction in the first plane and orthogonal to the first direction, a first portion of each projected line reflects from the reflective film. A reflected image of each line includes the reflected first portion. Each reflected image has a luminance distribution defining a centerline of the reflected image. A distribution of an angle α between the centerlines of the reflected images and the second direction has a full width at half maximum of less than 3 degrees.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1B are schematic cross-sectional views of a glass laminate and a light source;

FIG. 2 is a schematic cross-sectional view of an optical film;

FIG. 3A is a schematic illustration of projected lines;

FIG. 3B is a schematic illustration of reflected lines;

FIG. 3C is a schematic illustration of a distribution of angles between centerlines of the reflected lines of FIG. 3B and a specified direction;

FIG. 4 is a schematic illustration of a reflected luminance distribution of a reflected image;

FIGS. 5A-5C are schematic illustrations of illuminance distributions;

FIG. 6A is a schematic illustration of a luminance distribution of a primary reflected image portion and first and second ghost portions;

FIG. 6B is a schematic illustration of a reflected luminance distribution which includes contributions from the primary reflected image portion and the first and second ghost portions of FIG. 6A;

FIG. 7 is a schematic cross-sectional view of a glass laminate having opposed major surfaces defining an angle δ therebetween;

FIG. 8 is a schematic front view of a windshield;

FIGS. 9-12 are luminance distributions across reflected images of projected lines;

FIGS. 13A-13C are reflected images of pluralities of projected lines; and

FIG. 14 is a plot of a distribution of centerlines of reflected lines.

DETAILED DESCRIPTION

In the following description, reference is made to the accompanying drawings that form a part hereof and in which various embodiments are shown by way of illustration. The drawings are not necessarily to scale. It is to be understood that other embodiments are contemplated and may be made without departing from the scope or spirit of the present

description. The following detailed description, therefore, is not to be taken in a limiting sense.

Head-up displays typically include a display or projector which projects an image onto a windshield or a combiner which reflects the projected image to a viewer. In some cases, the windshield is a glass laminate which includes a reflective film between two glass layers for reflecting the projected image. Ghost images reflected from outer surfaces of the glass laminate can degrade the image quality of the reflected image. In some cases, a glass laminate has a wedge design that provides a difference in slope between the reflective film and at least one of the outer surfaces of the glass laminate. The difference in slope can be selected to shift the ghost image onto the image reflected by the film so that the ghost does not substantially degrade the sharpness of the reflected image. However, such a wedge design is often not preferred in many embodiments due, at least in part, to the difficulty of providing a desired slope difference in a cost-effective manufacturing process.

According to some embodiments of the present description, it has been found that utilizing a reflective film asymmetrically disposed between glass layers having substantially parallel outermost major surfaces can provide improved perceived image quality by shifting the ghost image that reflects from the front major surface so that it is closer to the primary reflected image. In some embodiments, at least one ghost image substantially overlaps with an image reflected from the reflective film. Traditionally, a relatively thick layer (e.g., 0.76 mm) of polyvinyl butyral (PVB) has been used to laminate glass layers together in a windshield. In some embodiments, a thin (e.g., 50 microns or less) adhesive layer is used to laminate the reflective film to the glass layer facing the projector and a thick (e.g., 700 microns or more) adhesive layer is used to laminate the reflective film to the opposite glass layer. This has been found to sufficiently shift the ghost image such that it is closer to, or substantially overlaps with, the primary reflected image and so does not substantially degrade the sharpness of the reflected image.

The thinner adhesive layer may be a traditional acrylate-based optically clear adhesive (OCA), for example, instead of a PVB layer commonly used in windshield glass laminates. A windshield glass laminate is sometimes included for its improved impact resistance compared to using a single glass layer. For example, one layer can hold glass fragments in place when an object impacts and cracks the other layer. It has been found that using a layer of OCA as the thin adhesive layer and a layer of PVB as the thick adhesive layer provides an impact resistance comparable to a traditional windshield glass laminate. In particular, in some embodiments, when a 5-pound steel ball is dropped on the glass laminate from 10 feet onto the glass layer adjacent the thicker adhesive layer, the ball is stopped by the laminate and no glass shards are separated from the glass laminate.

Another advantage of the glass laminates according to some embodiments of the present description is improved fidelity of the reflected image. Utilizing a reflective film between glass layers and using traditional windshield adhesive layers can result in a reduced flatness of the reflective film and this can result in a waviness when a line is projected onto the glass laminate, for example. It has been found that using a thinner adhesive layer on the side of the reflective film facing the projector reduces this waviness.

FIG. 1A is a schematic cross-sectional view of a glass laminate **100** and a light source **122**. The glass laminate **100** includes first and second glass layers **102** and **104** having substantially parallel outermost major surfaces **103** and **105** facing away from one another, and a reflective film **110**

having opposed first and second major surfaces **112** and **114** and disposed between the first and second glass layers **102** and **104** with the first and second major surfaces **112** and **114** facing the respective first and second glass layers **102** and **104**. In some embodiments, the reflective film **110** has an average reflectance for a first polarization state (e.g., polarization state **131** depicted in FIG. 1B which is a p-polarization state in the illustrated embodiment) in a predetermined visible wavelength range at a predetermined angle of incidence of at least 15% (e.g., in a range of 15%-30%, or about 20%) and an average transmittance for an orthogonal second polarization state (e.g., polarization state **132** depicted in FIG. 1B which is an s-polarization state in the illustrated embodiment) in the predetermined visible wavelength range at the predetermined angle of incidence of at least 30%. In some embodiments, the reflective film **110** includes a plurality of alternating polymeric interference layers as described further elsewhere herein. The glass laminate **100** includes a first adhesive layer **117** disposed between and bonding together the first glass layer **102** and the reflective film **110**, and a second adhesive layer **119** disposed between and bonding together the second glass layer **104** and the reflective film **110**. The second adhesive layer **119** can optionally include an optically absorbing material **144** as described further elsewhere herein.

In some embodiments, the second adhesive layer **119** is thicker than the first adhesive layer **117** such that the first major surface **112** of the reflective film **110** is separated from the outermost major surface **103** of the first glass layer **102** by a distance d_1 , the second major surface **114** of the reflective film **110** is separated from the outermost major surface **105** of the second glass layer **104** by a distance d_2 , and $0.05 \leq d_1/d_2 \leq 0.9$. In some embodiments, $0.05 \leq d_1/d_2 \leq 0.8$, or $0.1 \leq d_1/d_2 \leq 0.8$, or $0.2 \leq d_1/d_2 \leq 0.7$. In some embodiments, the second adhesive layer **119** is at least 2, 3, 5, 10, 20, 50, 100, or 200 times thicker than the first adhesive layer **117**. In some embodiments, the first adhesive layer **117** has a thickness in a range of 1 micron to 100 microns and the second adhesive layer **119** has a thickness in a range of 100 microns to 1000 microns. In some embodiments, the first adhesive layer **117** has a thickness in a range of 1 micron to 50 microns and the second adhesive layer **119** has a thickness in a range of 700 microns to 1000 microns.

In some embodiments, the first and second glass layers **102** and **104** have a substantially same thickness. In this context, substantially same thickness means within 5% of one another. In some embodiments, the first glass layer **102** has a thickness in a range of 0.95 to 1.05, or 0.97 to 1.03, or 0.98 to 1.02 times a thickness of the second glass layer **104**. In some embodiments, the second glass layer **104** is thicker than the first glass layer **102**. In some embodiments, the second glass layer **104** is at least 1.2 times, or 1.5 times, or 1.8 times, or 2 times thicker than the first glass layer **102**. In some embodiments, the second glass layer **104** is no more than 4 times, or 3 times or 2.5 times thicker than the first glass layer **102**. Using a thinner first glass layer **102** positions a first ghost image closer to a primary reflected image but using a thicker first glass layer **102** (e.g., having a thickness similar to that of the second glass layer **104**) improves impact resistance. In some embodiments, the first glass layer **102** has a thickness less than 2.2 mm, or less than 2 mm, or less than 1.5 mm, or less than 1.2 mm. In some embodiments, the first glass layer **102** has a thickness greater than 0.6 mm, or greater than 0.8 mm.

In some embodiments, the light source **122** emits or projects an image of a line having a projected luminance distribution about a centerline of the projected line having a

full width at half maximum σ . The luminance distribution may be expressed as a function of the x-coordinate illustrated in FIG. 1A or in terms of an angle from a peak luminance direction or from a central ray **127** as schematically illustrated in FIG. 1B. Non-central rays **129a** and **129b** are also illustrated in FIG. 1B. Ray **129b** makes an angle φ with the central ray **127**. The luminance distribution can be expressed in terms of the angle φ , where positive φ in FIG. 1B corresponds to positive x-coordinate in FIG. 1A. The luminance distribution can be determined using a detector having an input aperture in a plane perpendicular to a central ray reflected from the reflective film **110** (e.g., the x-y plane referring to the x-y-z coordinate system of FIG. 1A). Suitable detectors include the PROMETRIC 18 imaging colorimeter available from Radiant Vision Systems (Redmond, WA). The luminosity, which may also be referred to a brightness, can be defined as an integral over wavelengths of the radiance times the photopic luminosity function defined by the Commission Internationale de l'Éclairage (CIE) in the CIE 1931 color space. Any relations described herein regarding luminance or luminance distribution may also hold for radiance or radiance distribution or for intensity or intensity distribution.

In some embodiments, the light source **122** projects polarized light having a first polarization state **131**. An ambient light ray **133** having a second polarization state **132** is illustrated in FIG. 1B as being transmitted through reflective film **110** which may be a reflective polarizer. The light source **122** may be or include a display such as a liquid crystal display (LCD) or an organic light emitting diode (OLED) display. In some embodiments, various optical components (e.g., curved mirror(s) and/or optical lens(es)) are included in the light source **122** to provide the desired light output to the glass laminate **100**.

FIG. 2 is a schematic cross-sectional view of a reflective film **210**, which may correspond to reflective film **110**. Reflective film **210** includes a plurality of alternating polymeric interference layers **241** and **242**. In the illustrated embodiments, the plurality of alternating polymeric interference layers **241** and **242** is disposed on an optional skin layer **240**. In some embodiments, a second skin layer is disposed adjacent the plurality of alternating polymeric interference layers **241** and **242** opposite the skin layer **240**. The skin layer **240** may optionally include optically absorbing material **244**. Optically absorbing material **244** may be dyes, pigments, or a combination thereof which may be dispersed in a polymeric material of the skin layer **240**. In some embodiments, at least one of the interference layers **241** or **242** is oriented along a first direction (e.g., x1-direction), and the optically absorbing material **244** is or includes a dichroic dye at least partially oriented along the first direction. Any of these optically absorbing materials may optionally be included in second adhesive layer **119** instead of or in addition to be including in the skin layer **240**. The optically absorbing material may be included to reduce the brightness of a ghost image reflected from the outermost major surface **105** as described further elsewhere herein.

Interference layers reflect and transmit light primarily by optical interference. Reflecting and transmitting light primarily by optical interference means that the reflectance and transmittance of the interference layers can be reasonably described by optical interference or reasonably accurately modeled as resulting from optical interference. Adjacent pairs of interference layers having differing refractive indices reflect light by optical interference when the pair has a combined optical thickness (refractive index times physical thickness) of $\frac{1}{2}$ the wavelength of the light. Interference

layers typically have a physical thickness of less than 250 nm or less than 200 nm. Skin layers are typically noninterference layers which have an optical thickness too large to reflect and transmit light primarily by optical interference and typically have a physical thickness of greater than 1 micron or greater than 2 microns. The reflective film **210** can include many more interference layers than schematically illustrated in FIG. 2. For example, the reflective film **210** can include between 50 and 800 interference layers.

Suitable materials for the alternating interference layers **241** and **242** and for the skin layer **240** include, for example, polyethylene naphthalate (PEN), copolymers containing PEN and polyesters (e.g., polyethylene terephthalate (PET) or dibenzoic acid), glycol modified polyethylene terephthalate (PETg), polycarbonate (PC), poly(methyl methacrylate) (PMMA), or blends of these classes of materials.

Exemplary reflective films composed of polymer materials may be fabricated using coextruding, casting, and orienting processes. Methods of making such films are described in U.S. Pat. No. 5,882,774 (Jonza et al.) "Optical Film", U.S. Pat. No. 6,179,948 (Merrill et al.) "Optical Film and Process for Manufacture Thereof", U.S. Pat. No. 6,783,349 (Neavin et al.) "Apparatus for Making Multilayer Optical Films", and patent application publication US 2011/0272849 (Neavin et al.) "Feedblock for Manufacturing Multilayer Polymeric Films". Useful reflective films for use in head-up displays are described in U.S. Pat. Appl. No. 2004/0135742 (Weber et al.).

The reflective film may be a partial mirror or a partial reflective polarizer, for example. In some embodiments, the reflective film is oriented primarily along the x1 direction and has a stronger reflectivity for a first polarization state having the electric field along the x1 direction and a lower reflectivity for a second polarization state having the electric field along the x2 direction, referring to the x1-x2-x3 coordinate system illustrated in FIG. 2.

In some embodiments, the reflective film **110** or **210** has an average reflectance for a first polarization state in a predetermined visible wavelength range at a predetermined angle of incidence of at least 15% and an average transmittance for an orthogonal second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence of at least 30%. The predetermined visible wavelength range may be the entire visible wavelength range (about 400 nm to about 700 nm) or a portion of the visible wavelength range. In some embodiments, the predetermined visible wavelength range extends at least from 450 nm to 650 nm. In some embodiments, the predetermined visible wavelength range extends from 400 nm to 700 nm. In some embodiments, the reflective film **110** or **210** is reflective in narrow bands corresponding to wavelengths transmitted by red, green, and blue subpixels of a display, for example. In this case, the predetermined wavelength range may be a disjoint union of a red range, a green range, and a blue range. This can allow the reflective film to be transmissive for both polarization states for wavelengths between the red range and the green range and between the green range and the blue range and so can increase the transparency of the reflective film for ambient light.

The predetermined angle of incidence may be the angle θ (see FIG. 1A) where a light source **122** is adapted to project onto the glass laminate. The predetermined angle of incidence and/or the angle θ may be in a range of 30 degrees to 85 degrees, or in a range of 50 degrees to 75 degrees, or in a range of 55 degrees to 70 degrees, or in a range of 55 degrees to 68 degrees, or in a range of 59 degrees to 68 degrees, or in a range of 55 degrees to 65 degrees, or in a

range of 62 degrees to 65 degrees, or the predetermined angle may be about 55 degrees (e.g., 50 to 60 degrees, or 51 to 59 degrees), about 62 degrees (e.g., 58 to 66 degrees, or 59 to 65 degrees) or about 65 degrees (e.g., 61 to 69 degrees, or 62 to 68 degrees), for example.

In some embodiments, the average reflectance of the reflective film **110** or **210** for the first polarization state in the predetermined visible wavelength range at the predetermined angle of incidence is at least 20%, or at least 50%, or at least 70%. In some embodiments, the average transmittance if the reflective film **110** or **210** for the second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence is at least 50%, or at least 70%.

The average reflectance and average transmittance in the predetermined wavelength range refers to the reflectance and transmittance averaged (unweighted) over wavelengths in the predetermined wavelength range. The reflectance and transmittance are determined for light incident on the reflective film in air, unless indicated differently.

In some embodiments, the reflective film includes absorbing material on one side of the film (e.g., in a skin layer) and not on the other or includes more absorbing material on one side than the other. In this case, the reflectance and transmittance are determined for light incident on the reflective film on the side of the film opposite the absorbing material or opposite the side that is more absorbing. In some embodiments, the reflective film **210** is disposed between the first and second glass layers **102** and **104** with the skin layer **240** facing the second glass layer **104** and with absorbing material included in the skin layer **240**. As described further elsewhere herein, this may be done to reduce the luminance of a ghost image reflected from the outermost major surface **105** of the second glass layer **104**.

When a reflective film is included in a glass laminate using PVB layers having thicknesses traditionally used in glass laminates of windshields, a distortion of an image reflected from the reflective film can occur due to a reduced flatness of the film. According to the present description, when a thin adhesive layer, such as a thin layer of optically clear adhesive (e.g., an optically clear adhesive (e.g., acrylate based) commonly used in optical components), is used in place of a PVB layer having a thickness traditionally used in windshield laminates, that this distortion can be substantially reduced.

FIG. 3A is a schematic illustration of a plurality of parallel lines **350** which can be projected by a light source **122** onto the glass laminate **100**. FIG. 3B is a schematic illustration of reflected image **352** of the plurality of parallel lines **350**. FIG. 3C is a schematic illustration of a distribution **356** of an angle α between centerlines **354** of the reflected images **352** and the y-direction (see FIG. 1A). The distribution **356** has a full width at half maximum **358** which may be less than 3 degrees, for example.

The light source **122** projects a light **120** onto the glass laminate **100**. Portions **124**, **126** and **128** of the light reflect from the glass laminate **100**. The projected light **120** may be a projected line or a plurality of projected lines, for example. The portions **124**, **126**, and **128** can refer to portions of a projected line or portions of a plurality of projected lines as will be clear from the context. In some embodiments, the projected line(s) is in the first polarization state (e.g., a p-polarization state). In other embodiments, the projected line(s) are unpolarized.

The term “parallel lines” should be understood to refer to straight lines that are parallel to one another unless indicated differently. The term “projected line” should be understood

to refer to a projected straight line unless indicated differently. However, the term “centerline” is used to refer to a curve or line which may or may not be a straight line (e.g., the centerlines may be curved and/or irregular).

In some embodiments, a glass laminate **100** includes first and second glass layers **102** and **104** having substantially parallel outermost major surfaces **103** and **105**; and a reflective film **110** or **210** including a plurality of alternating polymeric interference layers **241** and **242** and disposed between and adhered to the first and second glass layers **102** and **104** through respective first and second adhesive layers **117** and **119**, where the first adhesive layer **117** has a thickness no more than 0.6 times (or no more than 0.5 times, or no more than 0.4 times, or no more than 0.2 times, or no more than 0.1 times) a thickness of the second adhesive layer **119**, such that when a light source **122** projects a plurality of parallel lines **350** onto the outermost major surface **103** of the first glass layer **102** (and through the first glass layer to the reflective film) along a first direction (z' direction) making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal **134** to the glass laminate **100** so that the plurality of parallel lines **350** extend along a second direction (y direction) orthogonal to a first plane (x' - z' plane) defined by the first direction and the normal **134** and are spaced apart along a third direction (x' -direction) in the first plane and orthogonal to the first direction, a first portion **124** of each projected line reflects from the reflective film **110** or **210**, where a reflected image **352** of each line includes the reflected first portion **124**, each reflected image **352** has a luminance distribution defining a centerline **354** of the reflected image **352**, and a distribution **356** of an angle α between the centerlines **354** of the reflected images **352** and the second direction (y-direction) has a full width at half maximum **358** of less than 3 degrees. The distribution **356** can be obtained by determining the angle α between the centerline **354** and the second direction at a plurality of locations along each line to determine the overall distribution of α . The plurality of locations can be selected at uniform intervals along the second direction and the number of locations can be increased until a statistical measure of the distribution, such as the full width at half maximum **358**, converges. Related image analysis procedures which can be used to determine the distribution of the orientation of the centerline tangent angle α are described in “Experimental investigation of collagen waviness and orientation in the arterial adventitia using confocal laser scanning microscopy”, Rezakhanlou et al., *Biomech Model Mechanobiol*, 2012 March; 11 (3-4); 461-73; doi: 10.1007/s10237-011-0325-z. In some embodiments, the full width at half maximum **358** of the distribution **356** of the angle α is less than 2 degrees, or less than 1.5 degrees, or less than 1.2 degrees, or less than 1.1 degrees.

In some embodiments, the light source **122** is positioned within 2 m, 1.5 m, 1.2 m, or 1 m of the glass laminate **100**. The distance between the light source **122** and the glass laminate **100** is the distance along a central light ray from the light source **122** to the glass laminate **100** (e.g., the distance between the light source **122** and the glass laminate **100** along the light **120**).

FIG. 4 is a schematic illustration of a reflected luminance distribution **460** of a reflected image. The distribution may be expressed in terms of a lateral dimension (x-dimension) at a detector location or in terms of an angle from a peak luminance direction (see, e.g., the angle ϕ illustrated in FIG. 1B). The distribution can be determined over a length of the line and so a non-zero angle α (see FIG. 3B) to the y-direction can increase the width of the distribution.

In some embodiments, a glass laminate **100** includes first and second glass layers **102** and **104** having substantially parallel outermost major surfaces **103** and **105**; and a reflective film **110** or **210** including a plurality of alternating polymeric interference layers **241** and **242** and disposed asymmetrically between the outermost major surfaces **103** and **105** such that when a light source **122** positioned within 2 m of the glass laminate **100** projects a line **350a** onto the outermost major surface **103** of the first glass layer **102** (and through the first glass layer to the reflective film) along a first direction (z' direction) making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal **134** to the glass laminate **100** so that the line extends along a second direction (y direction) orthogonal to a first plane (x'-z' plane) defined by the first direction and the normal **134** and has a projected luminance distribution about a centerline of the projected line having a full width at half maximum σ of no more than 0.05 degrees, a first portion **124** of the projected line **350a** reflects from the reflective film and a second portion **126** of the projected line **350a** reflects from the outermost major surface **103** of the first glass layer **102**, a reflected image **352a** of the line including a primary reflected image portion **472** (portion under dotted line in FIG. 4) defined by the reflected first portion **124** and a first ghost portion **474** (a portion between the dotted line and the solid line in FIG. 4) defined by the reflected second portion **126**. The first ghost portion **474** substantially overlaps with the primary reflected image portion **472**.

In some embodiments, a third portion **128** of the projected line **350** reflects from the outermost major surface **105** of the second glass layer **104**, and the reflected image **352** of the line further includes a second ghost portion **476** (a portion between the dotted line and the solid line in FIG. 4) defined by the reflected third portion **138**, where the second ghost portion **476** substantially overlaps with the primary reflected image portion **472**.

In some embodiments, the reflected image **352a** has a reflected luminance distribution **460** having a maximum at a peak **462** of the reflected luminance distribution **460** and decreasing monotonically in at least one lateral direction (+x direction) away from the peak **462** to an edge **464** of the reflected image **352a**. The edge **464** can be taken to be where the luminance drops to 5% of the maximum luminance.

In some embodiments, the reflected image **352a** has a reflected luminance distribution **460**, where a contribution to the reflected luminance distribution **460** from the first ghost portion **474** is not separately resolvable from a contribution to the reflected luminance distribution **460** from the primary reflected image portion **472** in a plot of the reflected luminance distribution **460**. The contribution from the first ghost portion **474** is not separately resolvable from the contribution from the primary reflected image portion **472** when there are no features in the distribution **460** that can be attributed to the first ghost portion **474** without reference to the primary reflected image portion **472**. For example, there are no local maxima or inflection points that can be attributed to the first ghost portion **474**. The first ghost portion **474** can be determined once the primary reflected portion **472** is determined. The primary reflected portion **472** can be determined from known luminance distribution of the projected line which allows the reflected luminance distribution to be determined when no ghosts are present. In the illustrated embodiment, the second ghost portion **476** is separately resolvable from the contribution from the primary reflected image portion **472** due to the presence of a local maxima and an inflection point on the left-hand side of the distribution **460**.

In some embodiments, the full width at half maximum of the projected line is no more than 0.03 degrees, or no more than 0.02 degrees. In some embodiments, the reflected image has an angular distribution of luminance having a full width at half maximum of no more than 0.1 degrees, or no more than 0.07 degrees, or no more than 0.05 degrees.

A portion of the reflected image substantially overlaps with another portion of the reflected image if the luminance of the portion having a larger maximum luminance is at least as large as the luminance of the other portion at the position (angular or linear) of a quarter maximum of the other portion. This is schematically illustrated in FIGS. 5A-5C. In FIG. 5A, the luminance of the primary reflected image portion **572** is substantially less than the luminance **577** at a quarter maximum of the first ghost portion **574** at the position **579a** of the quarter maximum. The full width at quarter maximum **578** of the first ghost portion **574** is indicated. The position **579a** is the quarter maximum position closest to the primary reflected image portion **572**. In FIG. 5B, the luminance of the primary reflected image portion **572** is equal to the luminance **577** at a quarter maximum of the first ghost portion **574** at the position **579b** of the quarter maximum. In FIG. 5C, the luminance of the primary reflected image portion **572** is greater than the luminance **577** at a quarter maximum of the first ghost portion **574** at the position **579c** of the quarter maximum. In the case illustrated in FIG. 5C, the luminance of the primary reflected image portion **572** is greater than the luminance at a half maximum of the first ghost portion **574** at a position of the half maximum. The full width at half maximum **588** of the first ghost portion **574** is indicated in FIG. 5C. The first ghost portion **574** substantially overlaps with the primary reflected image portion **572** in the cases illustrated in FIGS. 5B and 5C, but not in the case illustrated in FIG. 5A. Overlap of the second ghost portion with the primary reflected image portion is defined similarly. In some embodiments where a portion of the reflected image is described as substantially overlapping with another portion of the reflected image, the luminance of the portion having a larger maximum luminance is at least as large as the luminance of the other portion at the position (angular or linear) of a half maximum of the other portion.

FIG. 6A schematically illustrates a luminance distribution of a primary reflected image portion **672** substantially overlapping with first and second ghost portions **674** and **676**. FIG. 6B schematically illustrates the reflected luminance distribution **660** which includes contributions from the primary reflected image portion **672** and the first and second ghost portions **674** and **676**. The dotted lines indicate locations of peaks in the primary reflected image portion **672** and the first and second ghost portions **674** and **676**. The vertical direction (direction along the dotted lines) represents luminance in arbitrary units and the horizontal direction represents angular or linear displacement.

In some embodiments, the glass laminate **100** includes optically absorbing material disposed between the first glass layer **102** and the outermost major surface **105** of the second glass layer **104**. In some embodiments, the optically absorbing material is disposed between the reflective film **110** and the outermost major surface **105** of the second glass layer **104**, or between alternating polymeric interference layers of the reflective film **110** and the outermost major surface **105** of the second glass layer **104**. In some embodiments, the second glass layer **104** is optically absorbing (e.g., having an optically absorbing band in the near infrared which extends into the red portion of the visible spectrum). As described further elsewhere herein, the optically absorbing material

can be included in a skin layer **240** or in an adhesive layer **119**, for example. The optically absorbing material may be included to reduce the brightness of the second ghost compared to the first ghost. In some embodiments, the second ghost portion **476** has a brightness less than a brightness of first ghost portion **474**. In some embodiments, the second ghost portion **476** has a brightness less than 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.05, or 0.01 times a brightness of first ghost portion **474**. The brightness of the first and second ghost portions are the peak values of the luminance distributions of the first and second ghost portions.

In some embodiments, the optically absorbing material has an absorbance depending on polarization. For example, in some embodiments, the reflective film has an average reflectance for a first polarization state in a predetermined visible wavelength range at a predetermined angle of incidence of at least 15% (or at least 20%, or at least 50% or at least 70%) and an average transmittance for an orthogonal second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence of at least 30% (or at least 50%, or at least 70%), and the optically absorbing material is optically absorptive for light having the first polarization state and substantially optically transmissive for light having the second polarization state (e.g., the absorbance for the second polarizations state may be less than 0.2, or less than 0.1 times the absorbance for the first polarization state).

As used herein, “substantially parallel” outermost major surfaces are sufficiently close to parallel that any deviation from parallel results in a shift in a relative position of the peaks of the first and second ghost portions of less than 10 percent. Substantially parallel outermost major surfaces may be parallel or nominally parallel. FIG. 7 is a schematic illustration of a glass laminate **700** having outermost major surfaces **703** and **705** defining an angle δ therebetween. In some embodiments, substantially parallel outermost major surfaces define an angle δ therebetween of less than 0.05, 0.03, 0.02, 0.015, 0.012, 0.11, 0.01, 0.009, 0.007, 0.005, 0.003, or 0.001 degrees. The angle δ is an angle between tangent planes at the opposing outermost major surfaces at a location on the glass laminate. In some embodiments, δ is in any of the above ranges for each location on the glass laminate or for each location over at least 80% or 90% of an area of the glass laminate. In some embodiments, the reflective film **110** is substantially parallel with the outermost major surface **103** in the sense that any deviation from parallel results in a shift in a relative position of the peaks of the first ghost and primary reflected image portions of less than 10 percent. Similarly, in some embodiments, the reflective film **110** is substantially parallel with the outermost major surface **105** in the sense that any deviation from parallel results in a shift in a relative position of the peaks of the second ghost and primary reflected image portions of less than 10 percent.

FIG. 8 is a schematic front view of a windshield **800** which may be or include the glass laminate **100**, for example. In some embodiments, the reflective film covers substantially the entire windshield **800** (e.g., at least 80% or at least 90% of a surface area of the windshield). In some embodiments, the reflective film and the first and second glass layers are substantially coextensive with one another (e.g., any of the first and second glass layers and the reflective film may cover at least 80% or at least 90% of a surface area of any other of the first and second glass layers and the reflective film).

The present application is related to U.S. Prov. Pat. Appl. No. 62/735,567, filed Sep. 24, 2018, which is hereby incorporated herein by reference in its entirety.

The following is a list of illustrative embodiments of the present description.

A first embodiment is a glass laminate comprising:
first and second glass layers having substantially parallel outermost major surfaces facing away from one another;

a reflective film having opposed first and second major surfaces and disposed between the first and second glass layers with the first and second major surfaces facing the respective first and second glass layers, the reflective film having an average reflectance for a first polarization state in a predetermined visible wavelength range at a predetermined angle of incidence of at least 15% and an average transmittance for an orthogonal second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence of at least 30%;

a first adhesive layer disposed between and bonding together the first glass layer and the reflective film; and
a second adhesive layer disposed between and bonding together the second glass layer and the reflective film, the second adhesive layer being thicker than the first adhesive layer such that the first major surface of the reflective film is separated from the outermost major surface of the first glass layer by a distance d_1 , the second major surface of the reflective film is separated from the outermost major surface of the second glass layer by a distance d_2 , and $0.05 \leq d_1/d_2 \leq 0.9$.

A second embodiment is the glass laminate of the first embodiment, wherein $0.05 \leq d_1/d_2 \leq 0.8$, or $0.1 \leq d_1/d_2 \leq 0.8$, or $0.2 \leq d_1/d_2 \leq 0.7$.

A third embodiment is the glass laminate of the first or second embodiments, wherein the second adhesive layer is at least 2, 3, 5, 10, 20, 50, 100, or 200 times thicker than the first adhesive layer.

A fourth embodiment is the glass laminate of any one of the first to third embodiments, wherein the first adhesive layer has a thickness in a range of 1 micron to 75 microns and the second adhesive layer has a thickness in a range of 300 microns to 1000 microns.

A fifth embodiment is the glass laminate of any one of the first to fourth embodiments, wherein the second glass layer is at least 1.5 times thicker than the first glass layer.

A sixth embodiment is the glass laminate of any one of the first to fifth embodiments, wherein the average reflectance of the reflective film for the first polarization state in the predetermined visible wavelength range at the predetermined angle of incidence is at least 20%, and wherein the average transmittance if the reflective film for the second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence is at least 50%.

A seventh embodiment is a glass laminate comprising:
first and second glass layers having substantially parallel outermost major surfaces; and

a reflective film comprising a plurality of alternating polymeric interference layers and disposed asymmetrically between the outermost major surfaces such that when a light source positioned within 2 m of the glass laminate projects a line onto the outermost major surface of the first glass layer along a first direction making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal to the glass laminate so that the line extends along a second direction

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orthogonal to a first plane defined by the first direction and the normal and has a projected luminance distribution about a centerline of the projected line having a full width at half maximum of no more than 0.05 degrees, a first portion of the projected line reflects from the reflective film and a second portion of the projected line reflects from the outermost major surface of the first glass layer, a reflected image of the line comprising a primary reflected image portion defined by the reflected first portion and a first ghost portion defined by the reflected second portion, the first ghost portion substantially overlapping with the primary reflected image portion.

An eighth embodiment is the glass laminate of the seventh embodiment, wherein a third portion of the projected line reflects from the outermost major surface of the second glass layer, the reflected image of the line further comprising a second ghost portion defined by the reflected third portion, the second ghost portion substantially overlapping with the primary reflected image portion.

A ninth embodiment is the glass laminate of the seventh or eighth embodiments, wherein the reflected image has a reflected luminance distribution having a maximum at a peak of the reflected luminance distribution and decreasing monotonically in at least one lateral direction away from the peak to an edge of the reflected image.

A tenth embodiment is the glass laminate of any one of the seventh to ninth embodiments, wherein the full width at half maximum of the projected line is no more than 0.03 degrees, and wherein the reflected image has an angular distribution of luminance having a full width at half maximum of no more than 0.1 degrees.

An eleventh embodiment is the glass laminate of any one of the seventh to tenth embodiments, wherein the second ghost portion has a brightness less than 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.05, or 0.01 times a brightness of first ghost portion.

A twelfth embodiment is a glass laminate comprising: first and second glass layers having substantially parallel outermost major surfaces; and

a reflective film comprising a plurality of alternating polymeric interference layers and disposed between and adhered to the first and second glass layers through respective first and second adhesive layers, the first adhesive layer having a thickness no more than 0.6 times a thickness of the second adhesive layer, such that when a light source projects a plurality of parallel lines onto the outermost major surface of the first glass layer along a first direction making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal to the glass laminate so that the plurality of parallel lines extend along a second direction orthogonal to a first plane defined by the first direction and the normal and are spaced apart along a third direction in the first plane and orthogonal to the first direction, a first portion of each projected line reflects from the reflective film, a reflected image of each line comprising the reflected first portion, each reflected image having a luminance distribution defining a centerline of the reflected image, a distribution of an angle α between the centerlines of the reflected images and the second direction having a full width at half maximum of less than 3 degrees.

A thirteenth embodiment is the glass laminate of the twelfth embodiment, wherein the reflective film has an average reflectance for a first polarization state in a predetermined visible wavelength range at a predetermined angle of incidence of at least 15% and an average transmittance for

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an orthogonal second polarization state in the predetermined visible wavelength range at the predetermined angle of incidence of at least 30%.

A fourteenth embodiment is the glass laminate of the twelfth or thirteenth embodiment, wherein the first adhesive layer has a thickness in a range of 1 micron to 75 microns and the second adhesive layer has a thickness in a range of 300 microns to 1000 microns.

A fifteenth embodiment is the glass laminate of any one of the twelfth to fourteenth embodiments, wherein the full width at half maximum of the distribution of the angle α is less than 2 degrees, or less than 1.5 degrees, or less than 1.2 degrees.

EXAMPLES

Reflective Film WCF

A reflective film, referred to as windshield combiner film (WCF), was made by extruding and uniaxially orienting 275 alternating polymer layers plus two outermost skin layers as generally described in U.S. Pat. No. 6,827,886 (Neavin et al.). The alternating polymer layers were oriented PET as the higher index layers and crystalline PETg as the lower index layers. The layer thicknesses were selected to produce reflectivity throughout the visible wavelength range of 400 nm to 700 nm. The film had an average reflectance in the visible range at an angle of incidence of 60 degrees of about 20% for p-polarized light and was substantially transmissive for s-polarized light.

Example 1

The reflective film WCF was laminated between first and second 2.1 mm thick glass layers with a 1 mil thick layer of 3M 8146 bonding the reflective film to the first glass layer and a 0.76 mm thick PVB layer (formed from two 0.38 mm thick PVB layers) bonding the reflective film to the second glass layer.

An Apple Inc. (Cupertino, CA) IPAD (first generation) was used to project a line onto the first glass layer of the glass laminate at an angle of incidence of about 65 degrees. The line image was 2 pixels wide. The reflected luminance distribution was determined using a PROMETRIC 18 imaging colorimeter available from Radiant Vision Systems (Redmond, WA) as a function of angle (e.g., the angle φ of FIG. 1B) from a central peak of the primary reflected image and is shown in FIG. 9. The primary reflected image had a full width at half maximum of about 0.02 degrees. A first ghost image was present at about 0.08 degrees from the primary reflected image and a second ghost image was present at about 0.11 degrees from the primary reflected image.

Example 2

Example 2 was prepared and tested as described for Example 1 except that the first and second glass layers were each 1 mm thick. The resulting reflected luminance distribution is shown in FIGS. 10-11 as a function of angle (e.g., the angle φ of FIG. 1B) and location (e.g., the x-coordinate of FIG. 1A), respectively. The first and second ghost images substantially overlap with the primary reflected image.

Comparative Example C1

Comparative Example C1 was prepared and tested as described for Example 1 except that the first adhesive layer

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was a 0.76 mm thick PVB layer. The resulting reflected luminance distribution is shown in FIG. 12 as a function of location (x-coordinate of FIG. 1A). The first and second ghost images has a sufficiently high luminance and were sufficiently displaced from the primary reflected image to cause objectionable loss of image fidelity.

Examples 3-4

Example 3 was prepared as described for Example 1 except that the first adhesive layer was a 0.38 mm thick PVB layer. Example 4 was prepared as described for Example 1 except that the first glass layer was 3.2 mm thick. Using a thicker glass layer is expected to affect the position of the ghost image but is expected to have a negligible effect on line waviness.

A plurality of parallel lines was projected onto the reflective film through the first glass layer of the glass laminate of Comparative Example C1, Example 3 and Example 4 at an angle of incidence on the glass laminate of about 55 degrees and the reflected image was analyzed using the PROMETRIC colorimeter. The reflected images for Comparative Example C1, Example 3 and Example 4 are shown in FIGS. 13A-13C, respectively. A centerline tangent to each line was determined at a sufficient number of locations along each line to determine a distribution of orientation of the centerline tangent using an image analysis procedure generally described in "Experimental investigation of collagen waviness and orientation in the arterial adventitia using confocal laser scanning microscopy", Rezakhaniha et al., *Biomech Model Mechanobiol*, 2012 March; 11 (3-4); 461-73; doi: 10.1007/s10237-011-0325-z. The resulting distribution is plotted in FIG. 14. The full width at half maximum for each distribution was determined and found to be 4.0 degrees, 1.9 degrees, and 1.0 degrees for Comparative Example C1, Example 3 and Example 4, respectively.

All references, patents, and patent applications referenced in the foregoing are hereby incorporated herein by reference in their entirety in a consistent manner. In the event of inconsistencies or contradictions between portions of the incorporated references and this application, the information in the preceding description shall control.

Descriptions for elements in figures should be understood to apply equally to corresponding elements in other figures, unless indicated otherwise. Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations can be substituted for the specific embodiments shown and described without departing from the scope of the present disclosure.

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This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

What is claimed is:

1. A glass laminate comprising:

first and second glass layers having substantially parallel outermost major surfaces; and

a reflective film comprising a plurality of alternating polymeric interference layers and disposed asymmetrically between the outermost major surfaces such that when a light source positioned within 2 m of the glass laminate projects a line onto the outermost major surface of the first glass layer along a first direction making an angle θ in a range of 30 degrees to 85 degrees with respect to a normal to the glass laminate so that the line extends along a second direction orthogonal to a first plane defined by the first direction and the normal and has a projected luminance distribution about a centerline of the projected line having a full width at half maximum of no more than 0.05 degrees, a first portion of the projected line reflects from the reflective film and a second portion of the projected line reflects from the outermost major surface of the first glass layer, a reflected image of the line comprising a primary reflected image portion defined by the reflected first portion and a first ghost portion defined by the reflected second portion, the first ghost portion substantially overlapping with the primary reflected image portion.

2. The glass laminate of claim 1, wherein a third portion of the projected line reflects from the outermost major surface of the second glass layer, the reflected image of the line further comprising a second ghost portion defined by the reflected third portion, the second ghost portion substantially overlapping with the primary reflected image portion.

3. The glass laminate of claim 1, wherein the reflected image has a reflected luminance distribution having a maximum at a peak of the reflected luminance distribution and decreasing monotonically in at least one lateral direction away from the peak to an edge of the reflected image.

4. The glass laminate of claim 1, wherein the full width at half maximum of the projected line is no more than 0.03 degrees, and wherein the reflected image has an angular distribution of luminance having a full width at half maximum of no more than 0.1 degrees.

5. The glass laminate of claim 1, wherein the second ghost portion has a brightness less than 0.6 times a brightness of first ghost portion.

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